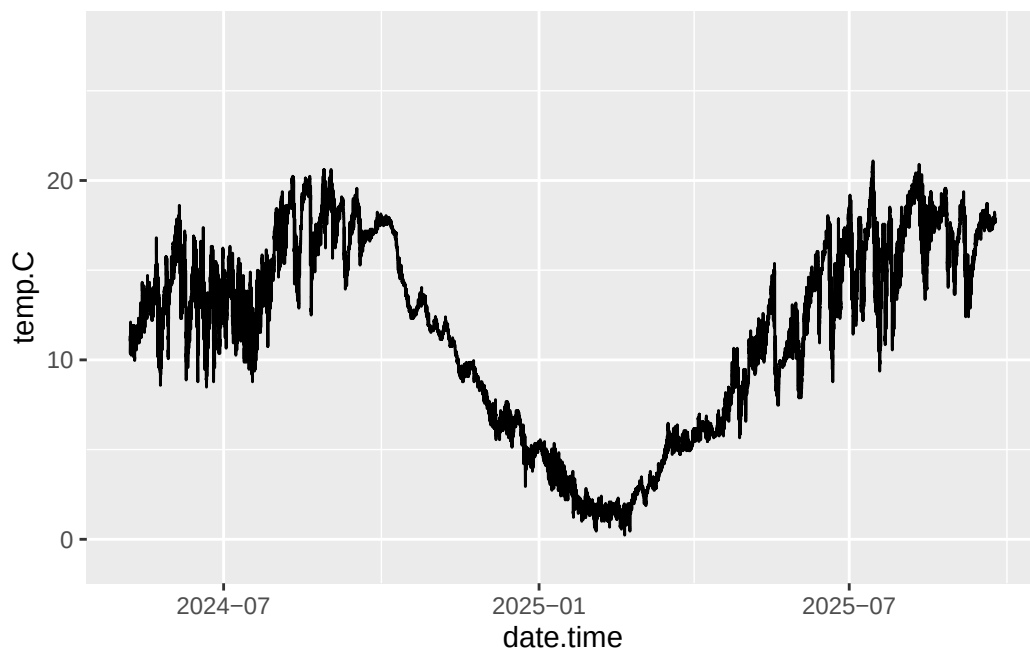


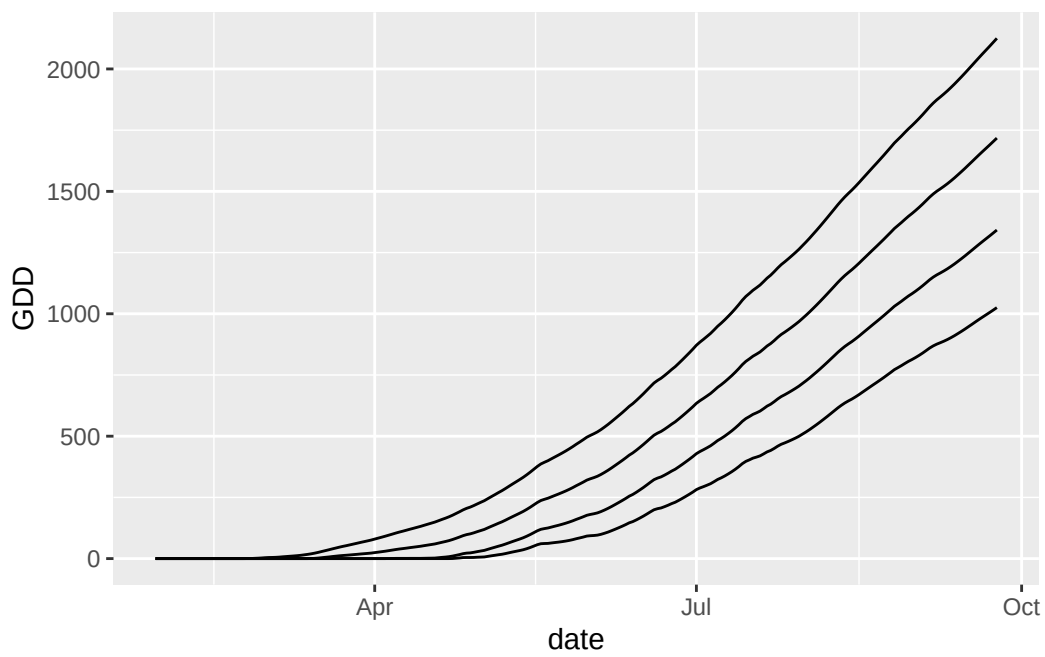
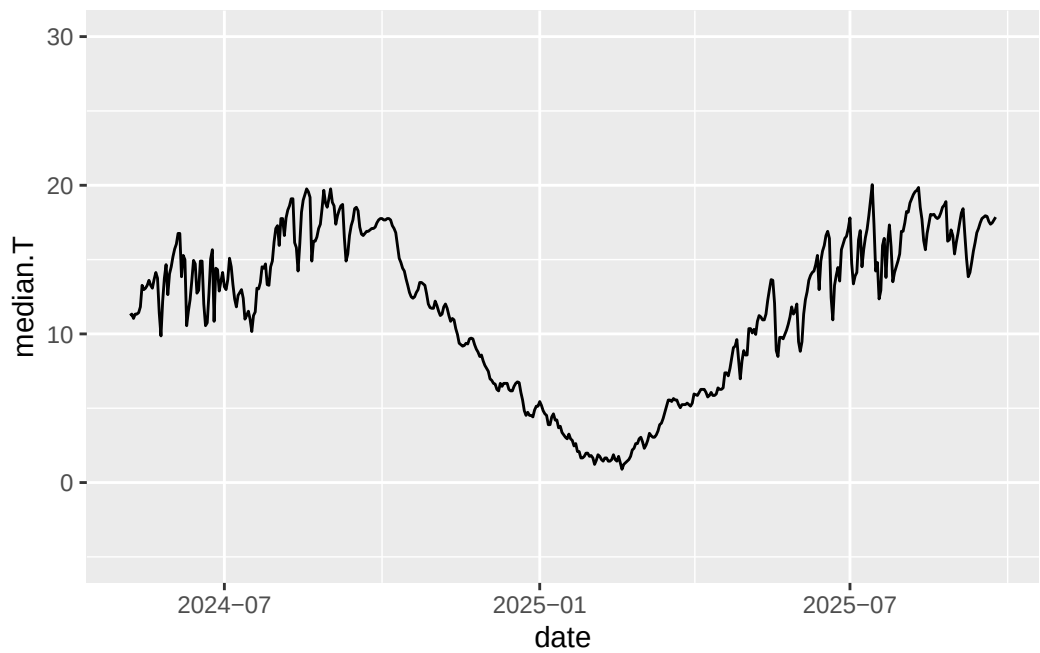
HOBO_SatTemp_Corr_2025

Aaron Shavitz, Forest Schenck

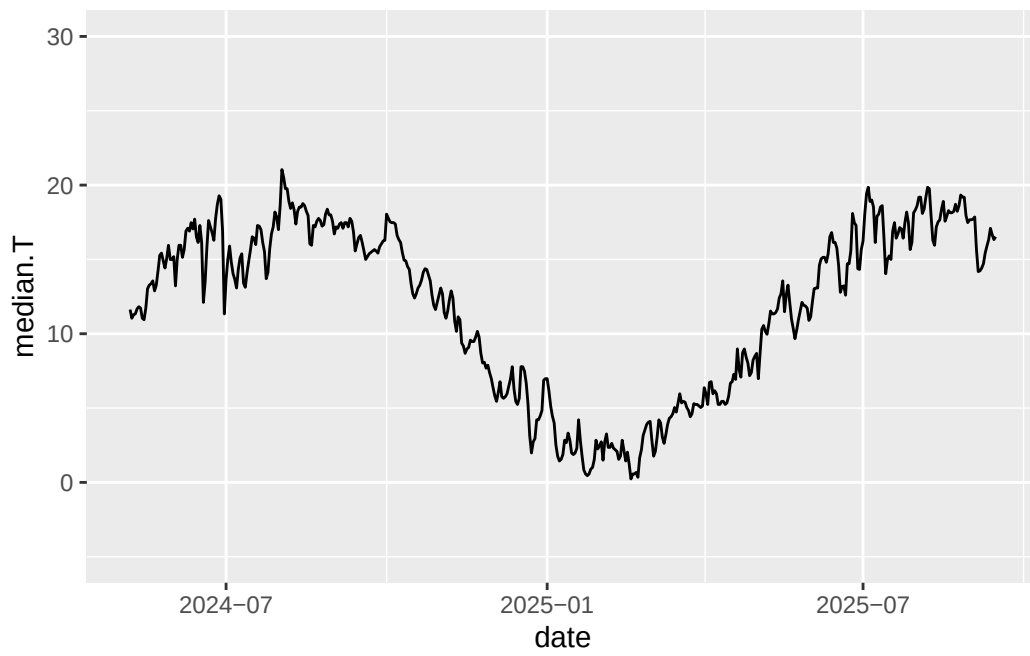
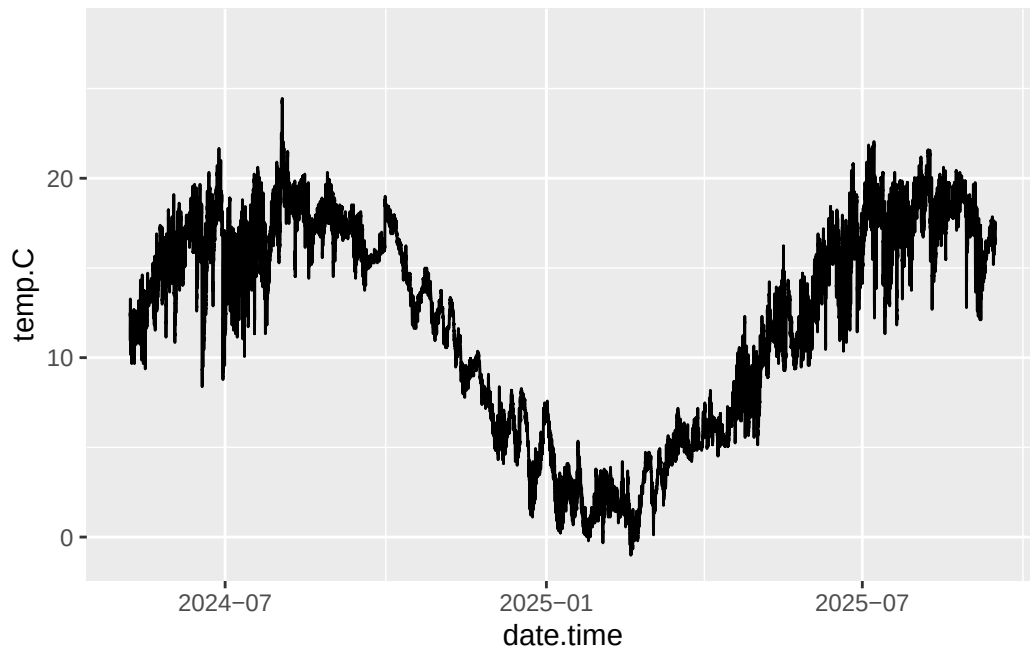
Data Carpentry Hobos

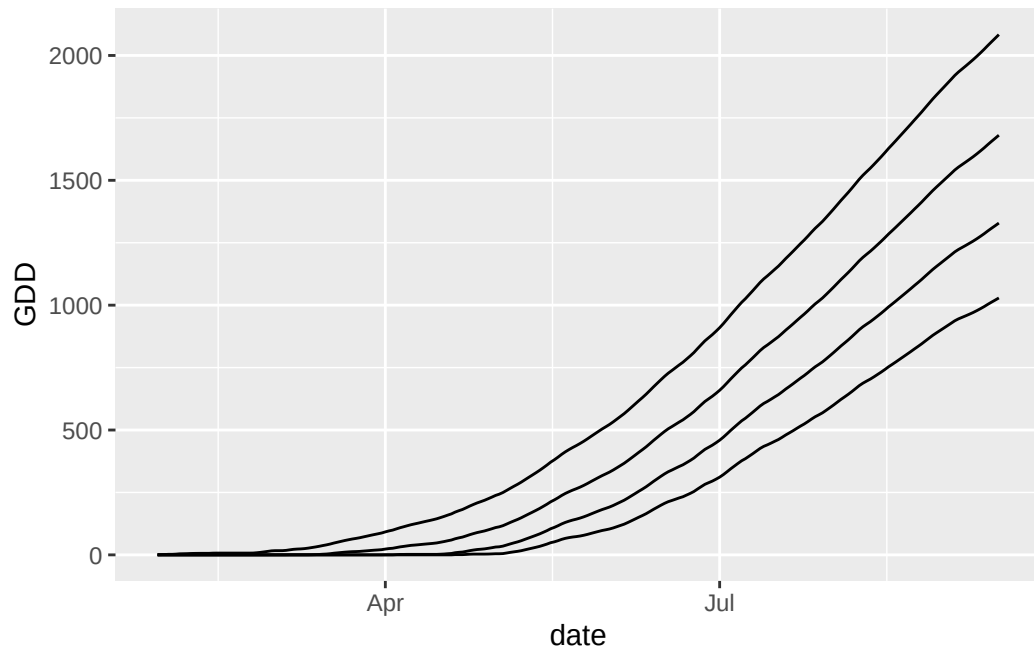
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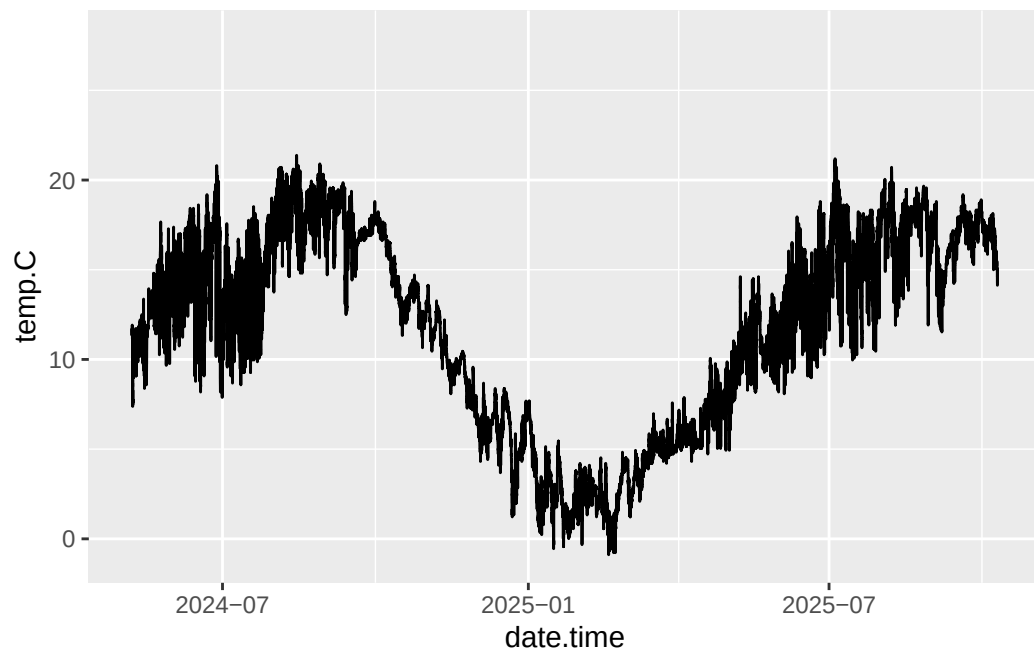


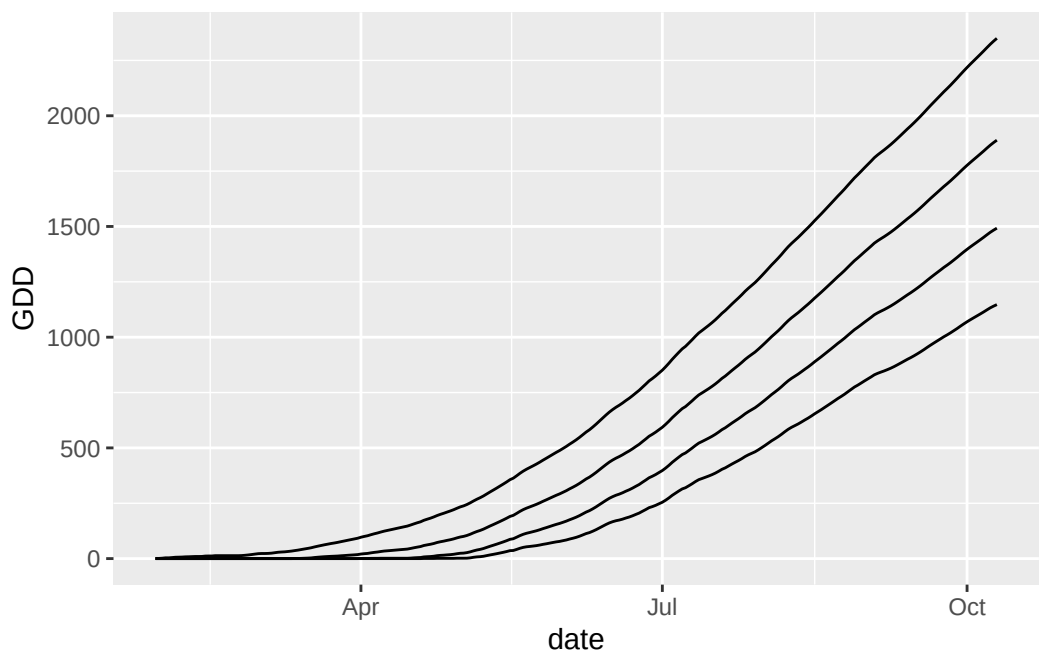
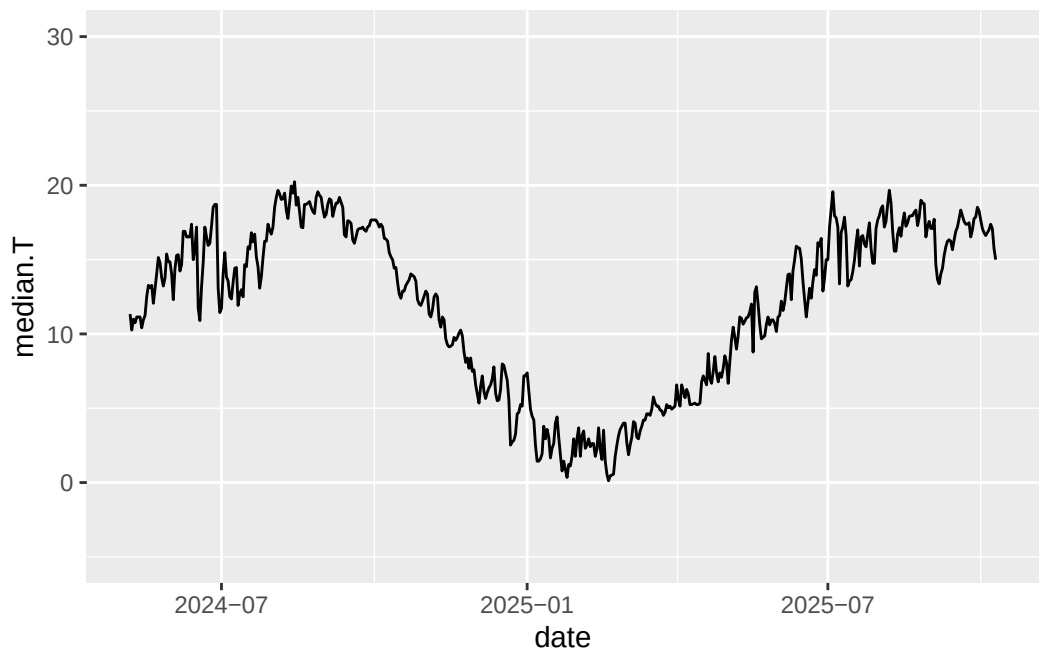
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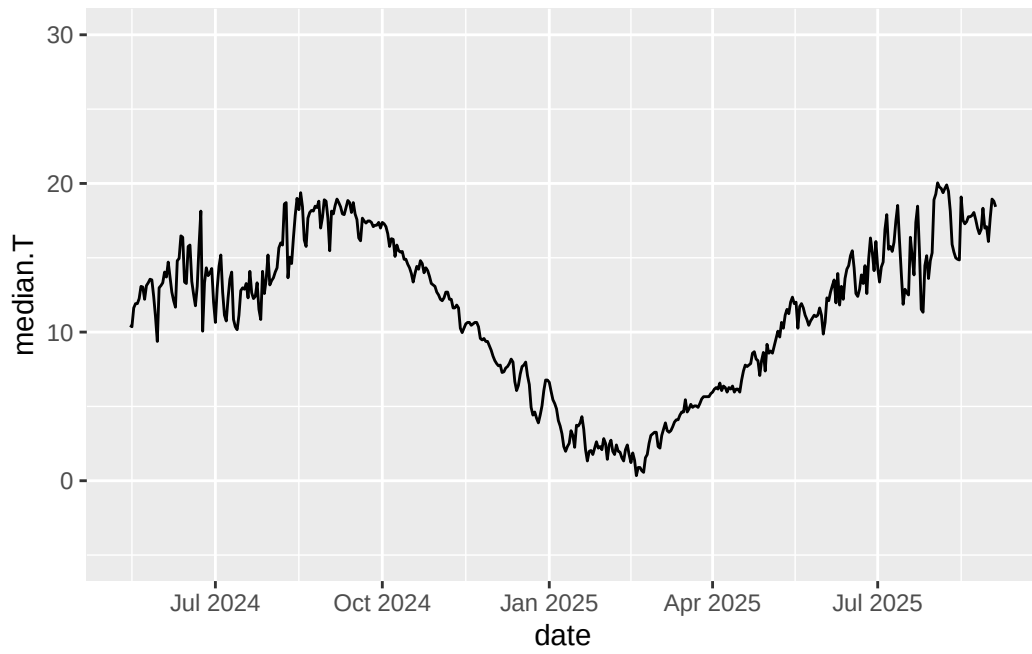
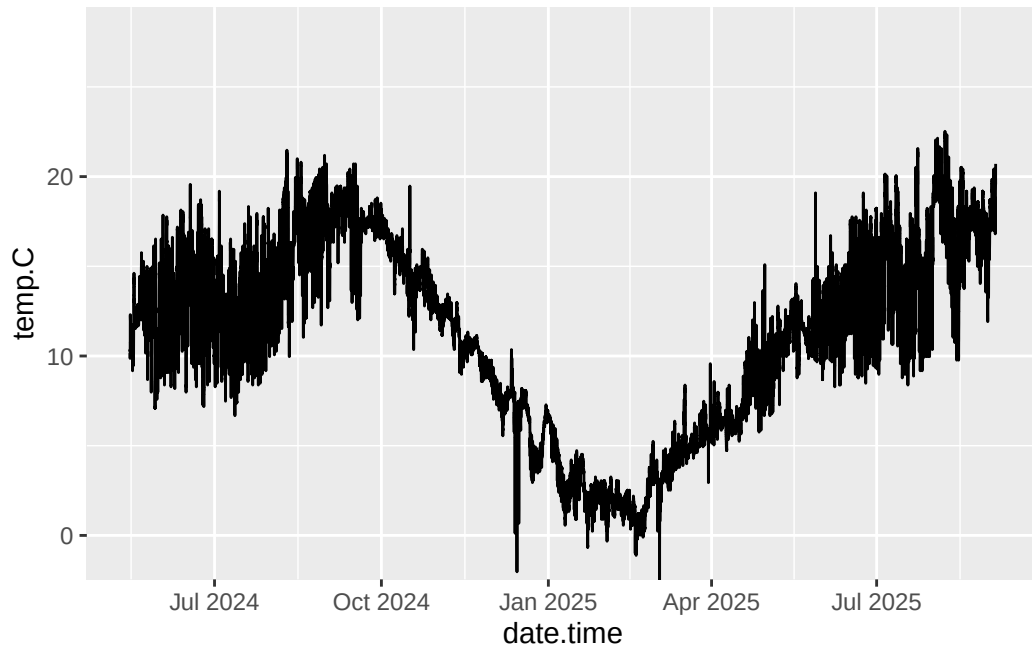


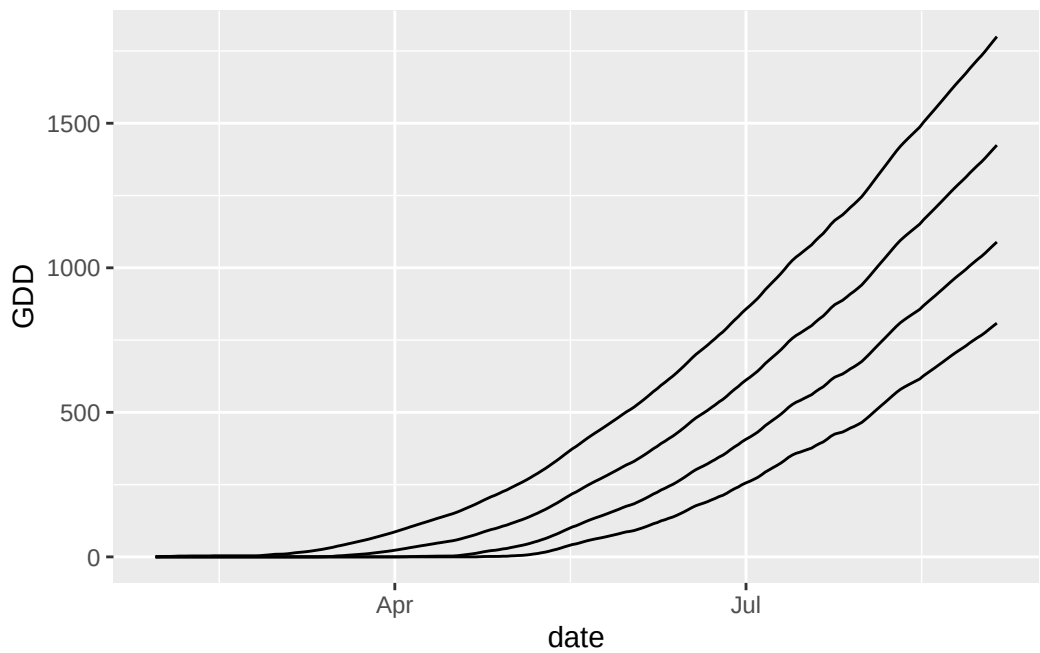
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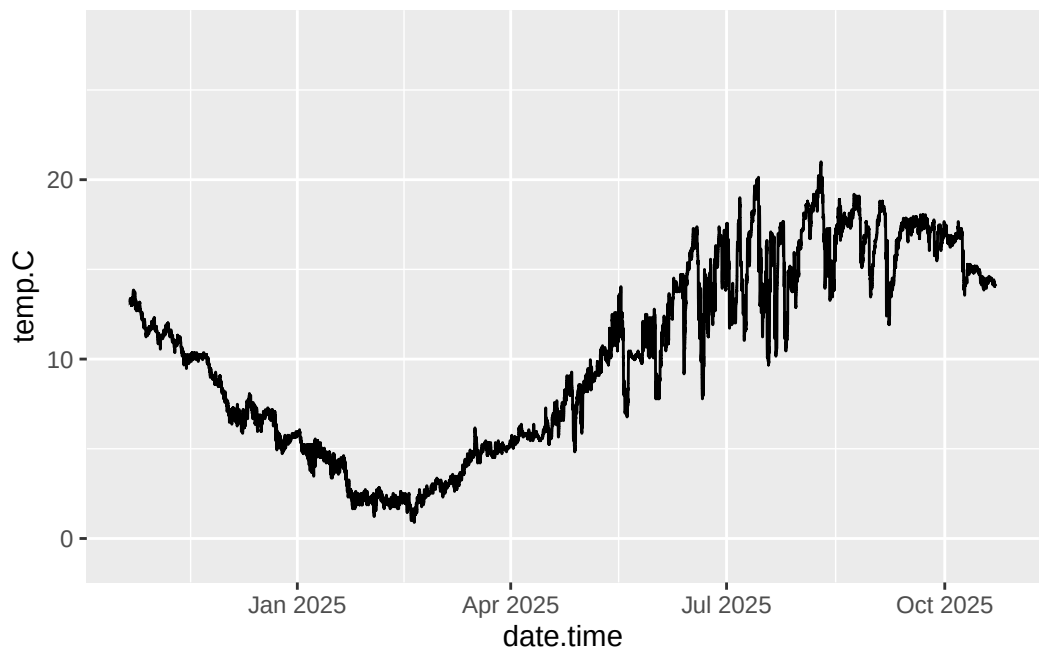


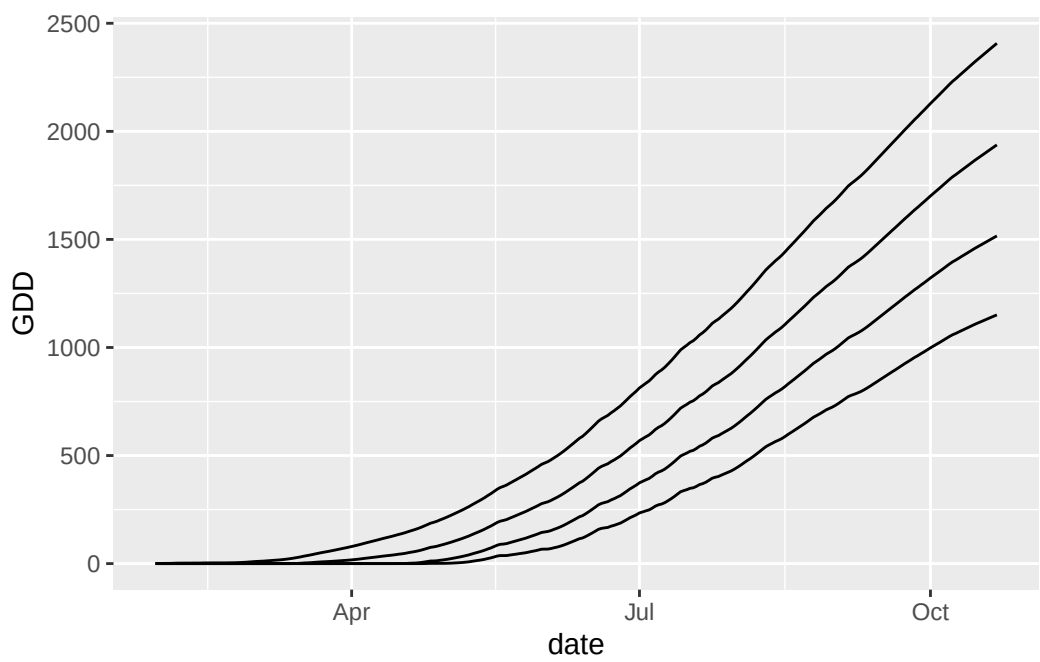
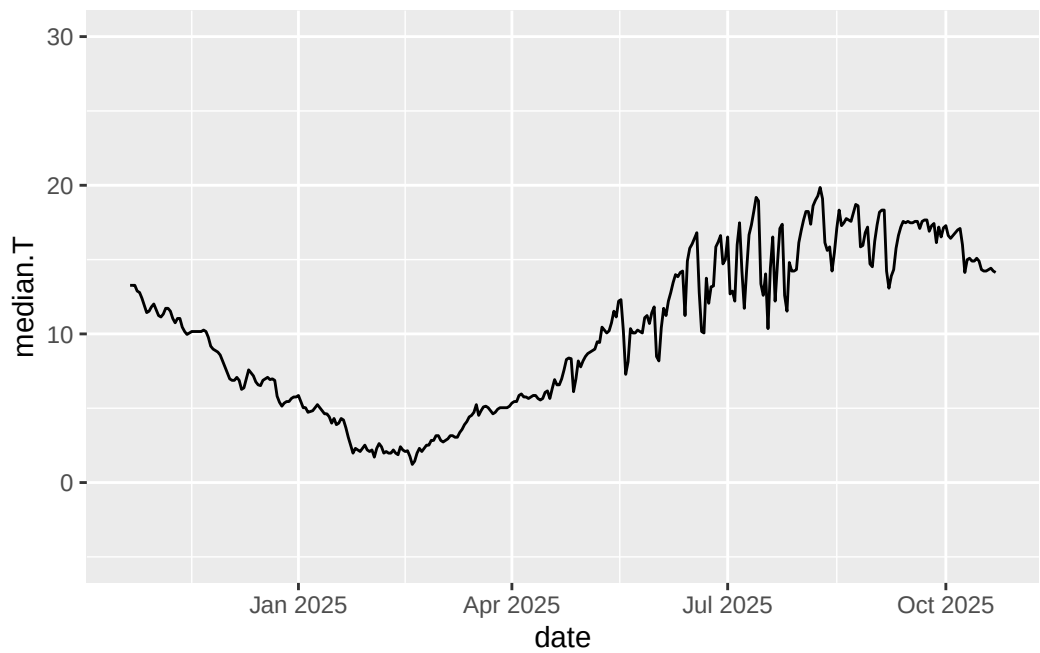
SH*



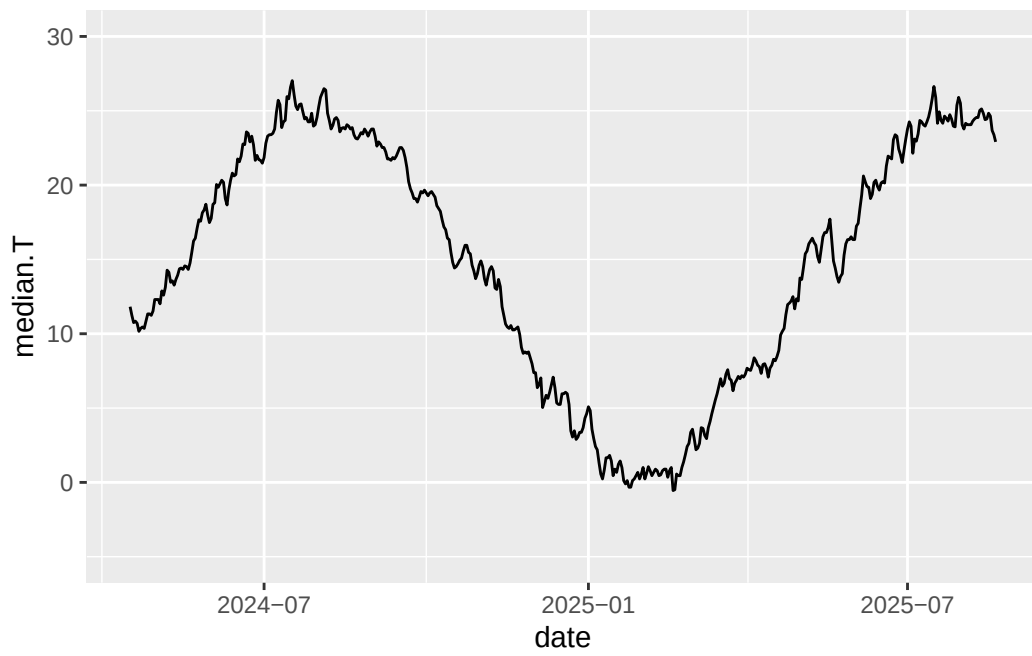
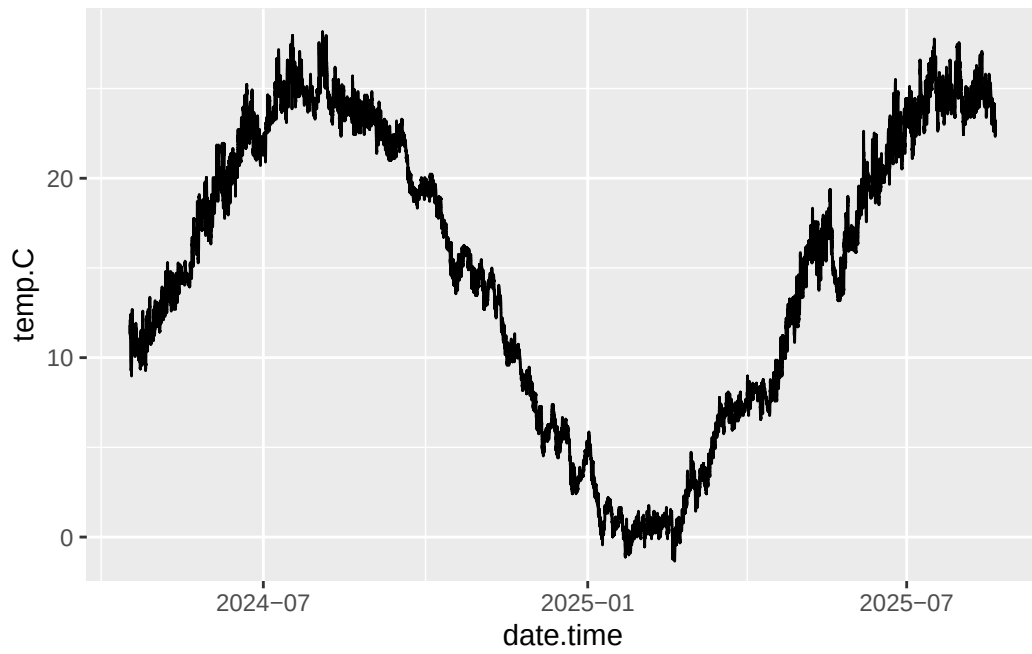


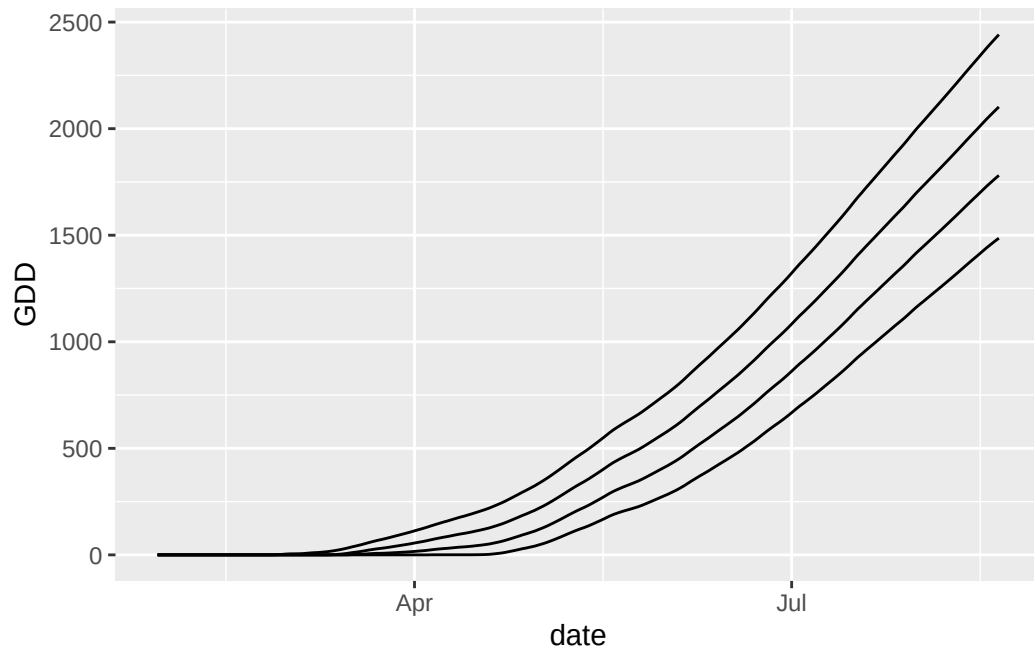
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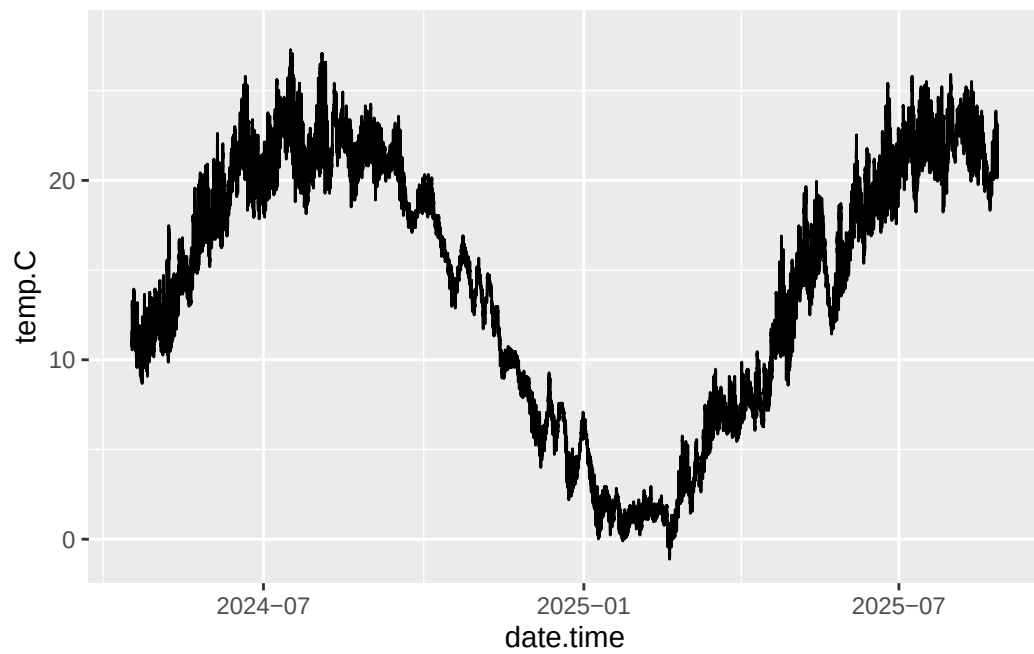


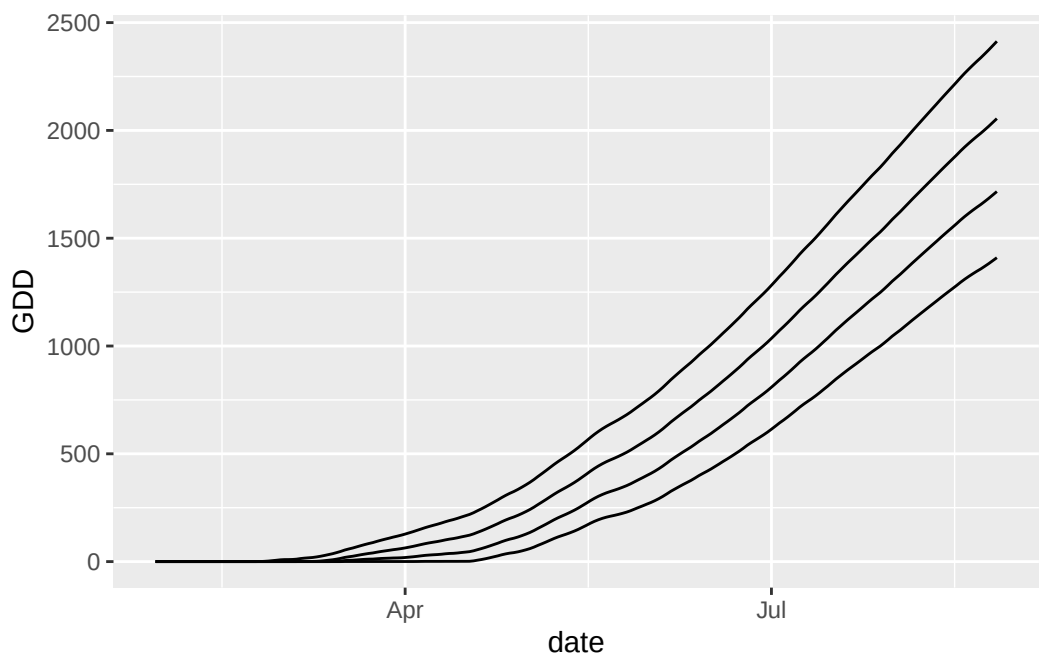
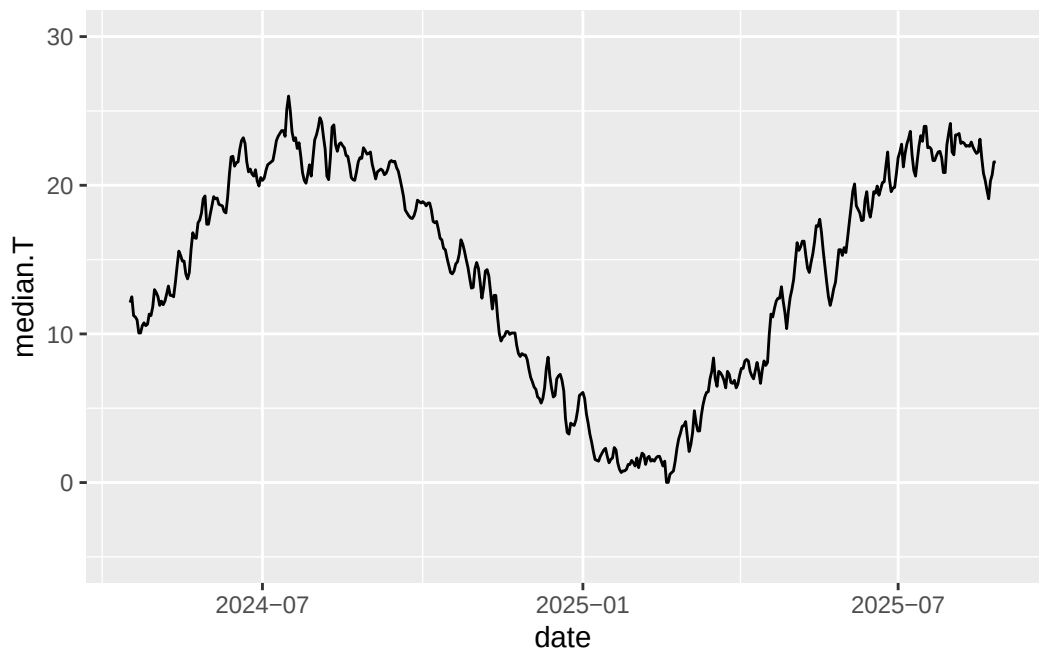
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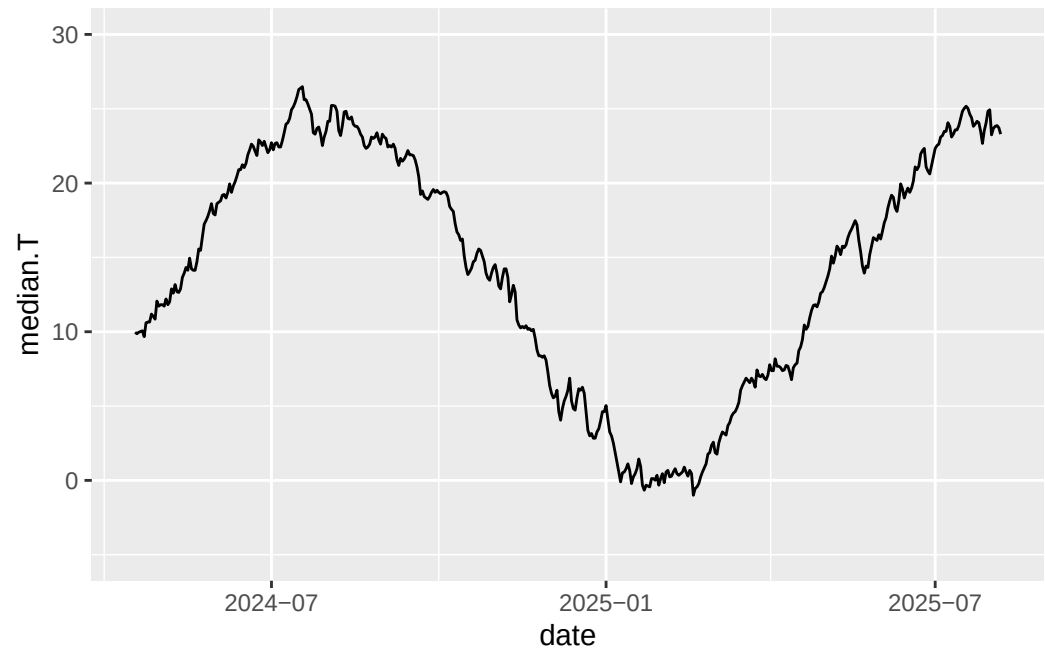
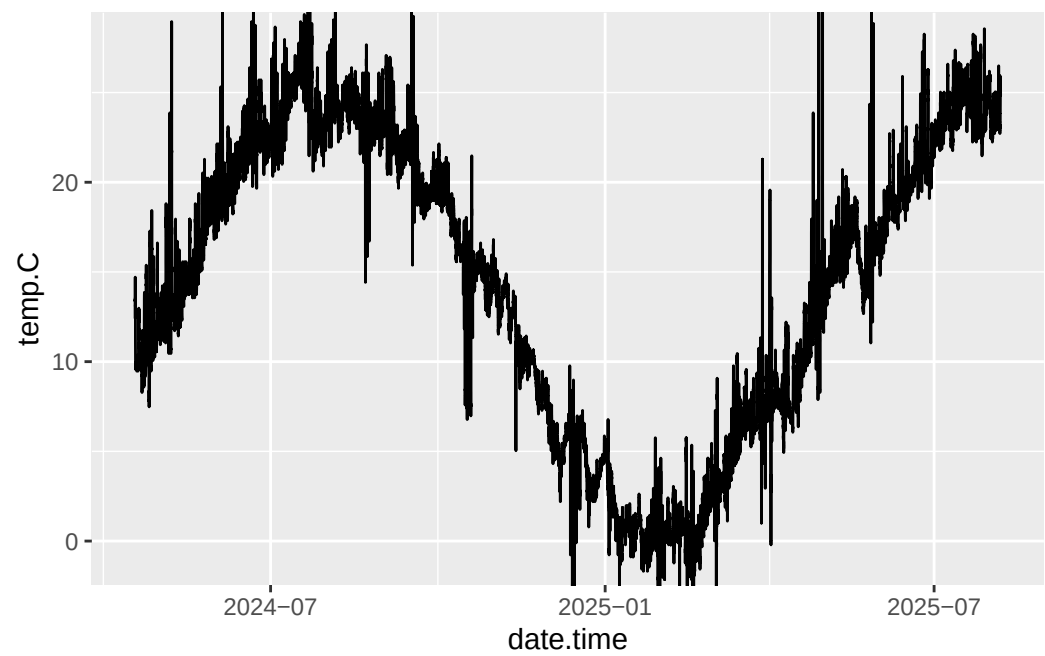


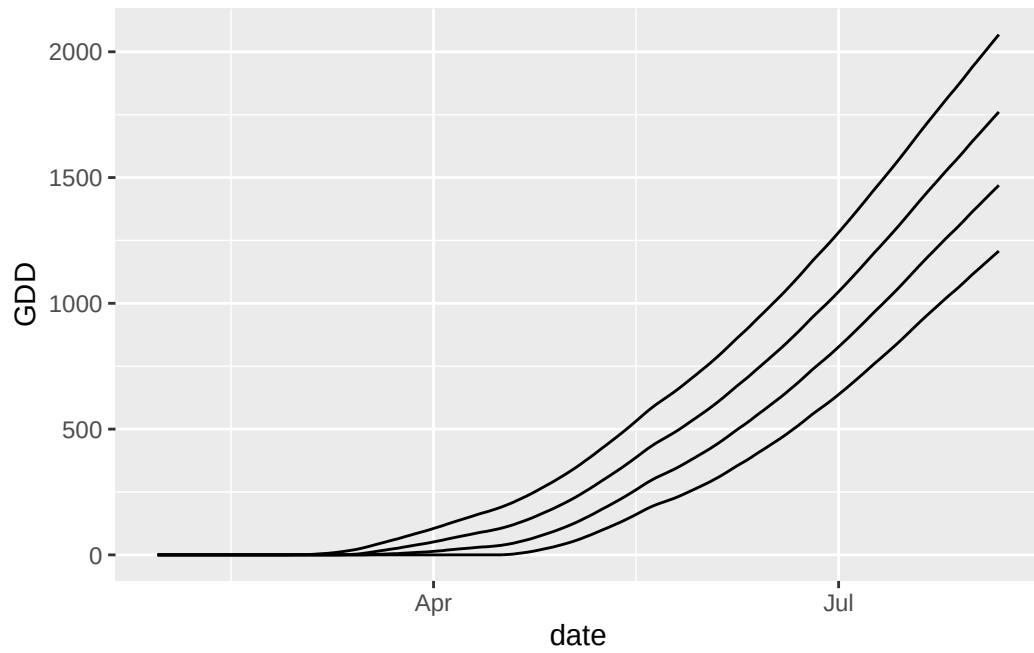
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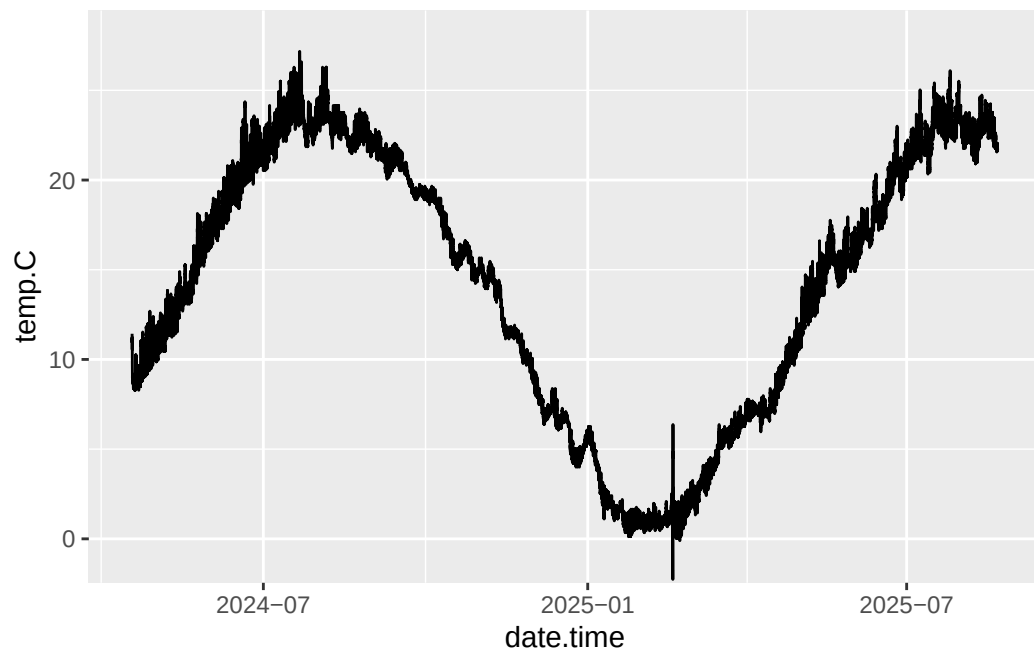


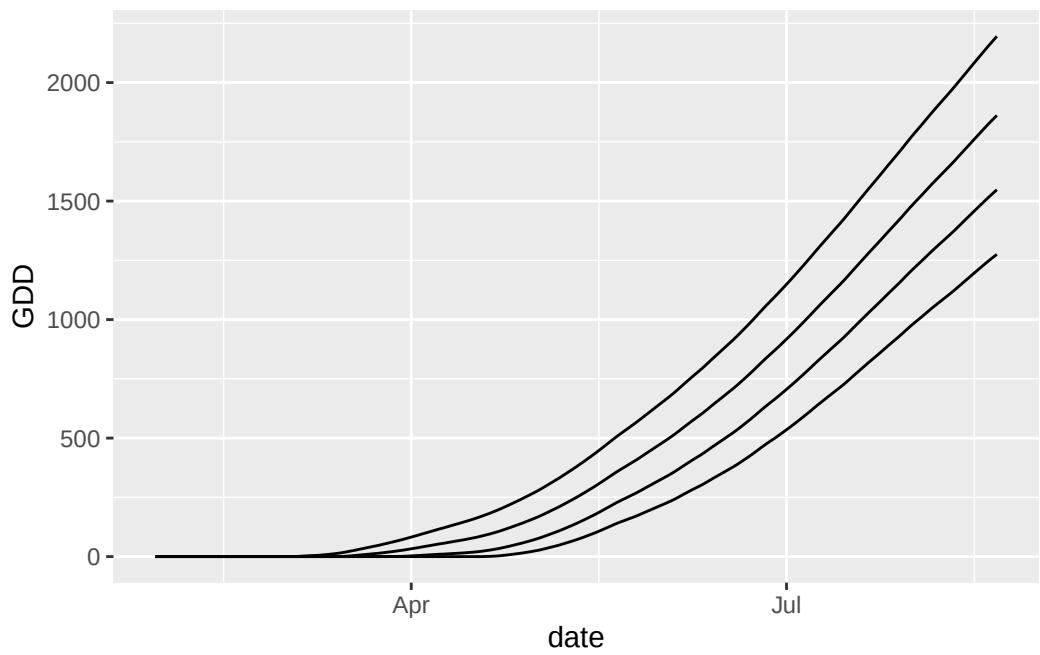
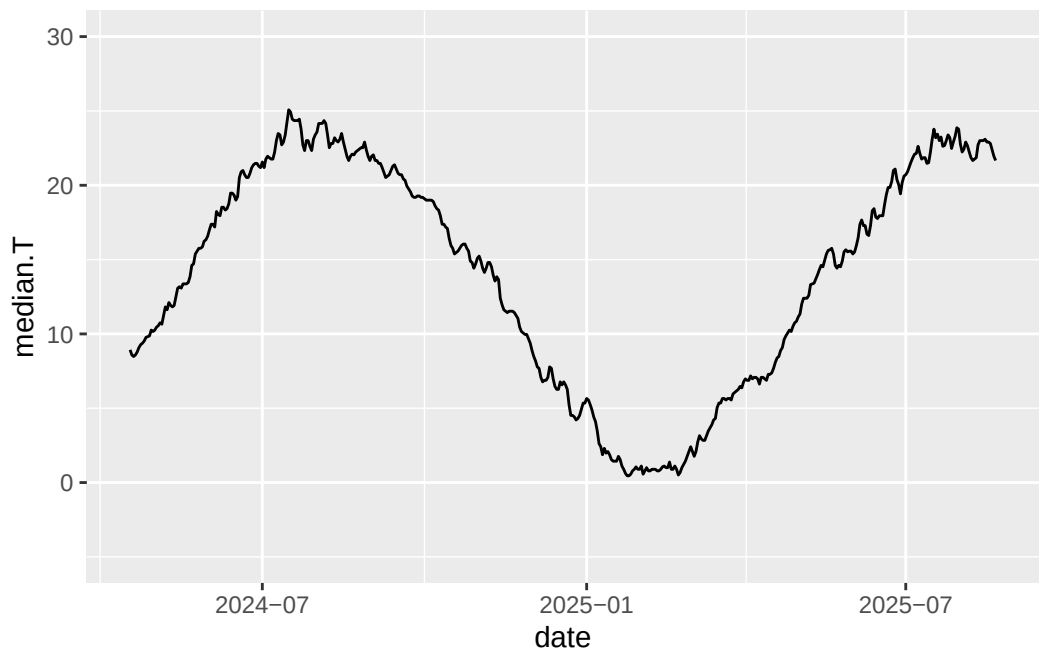
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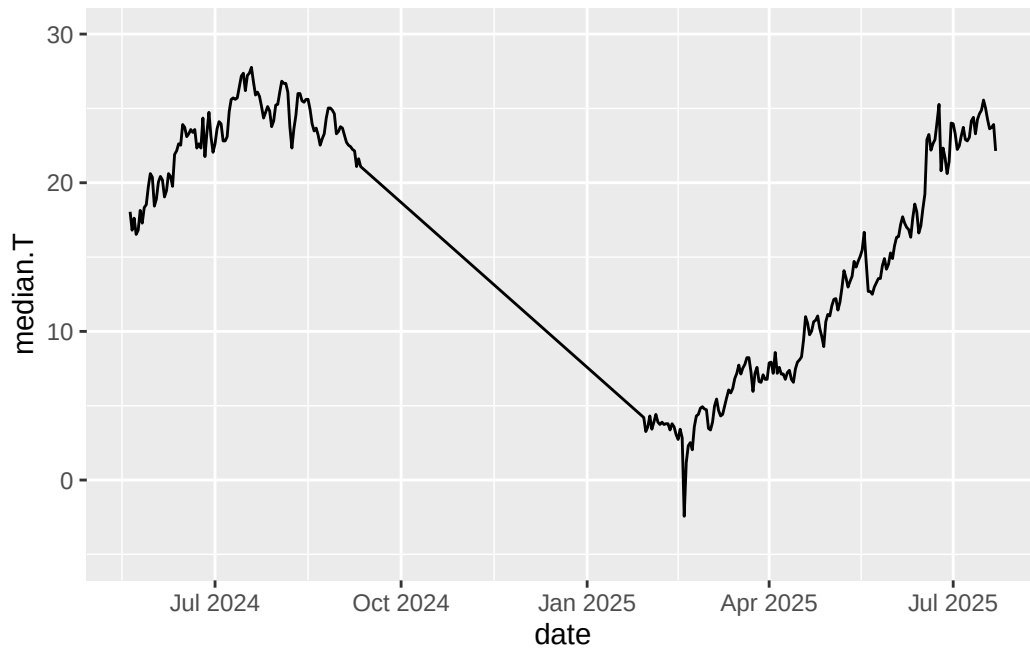
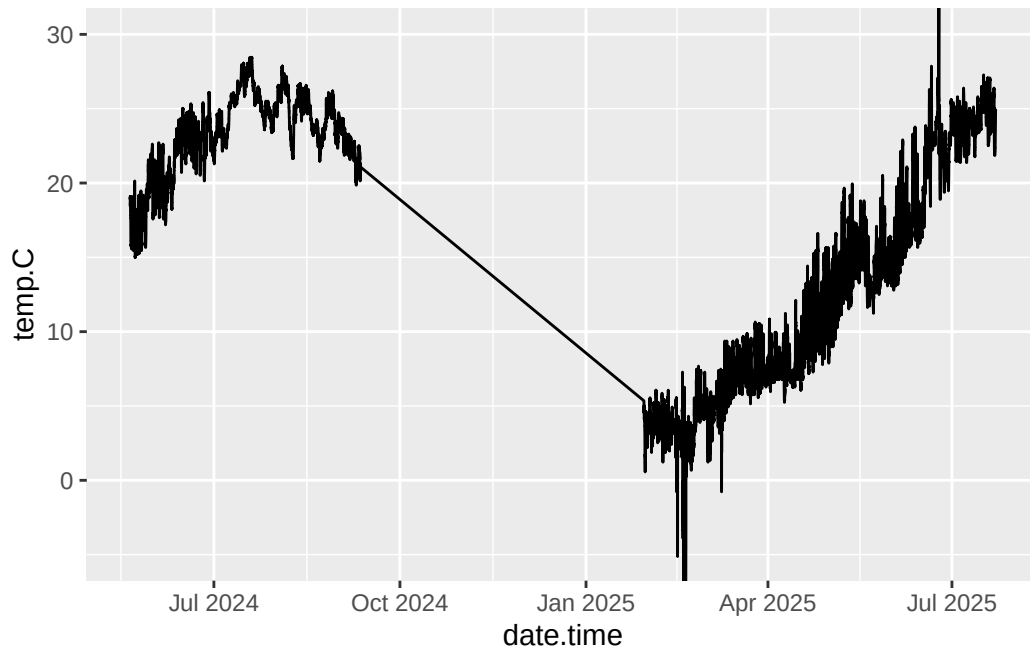


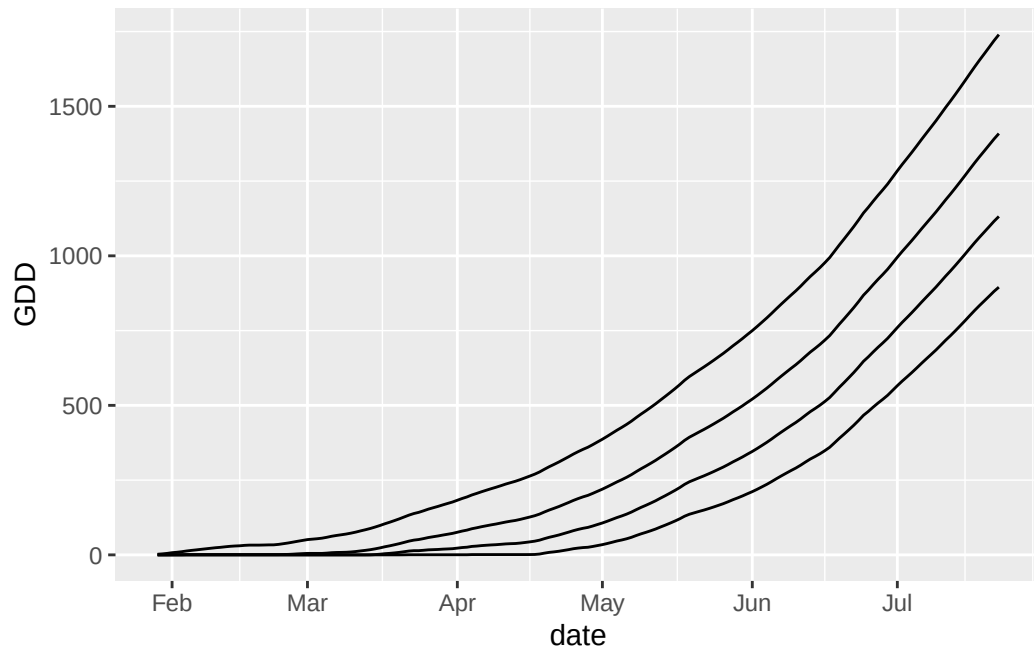
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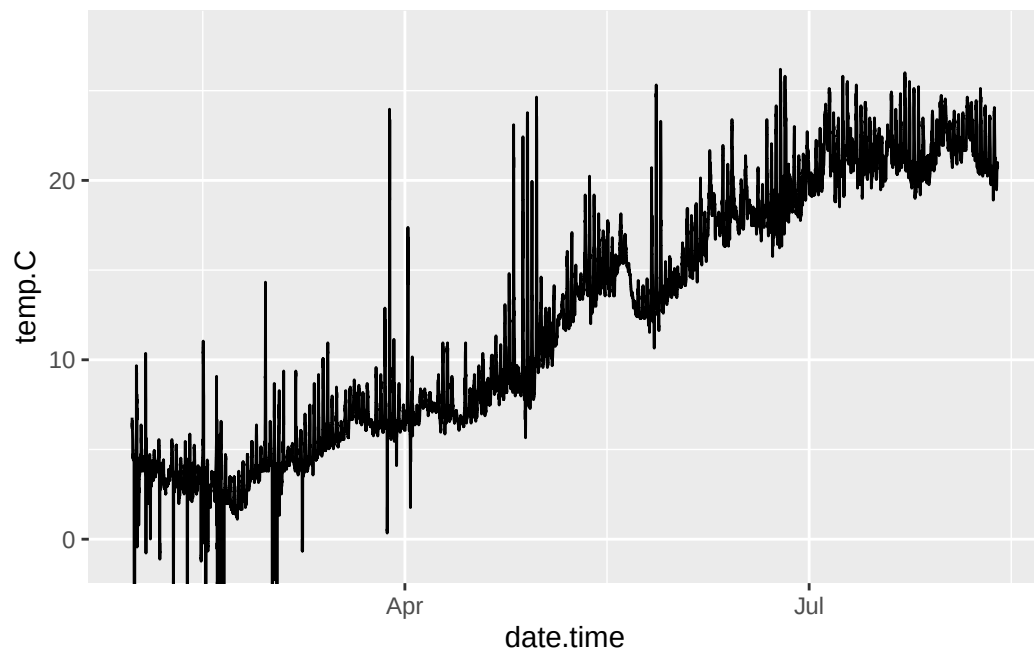


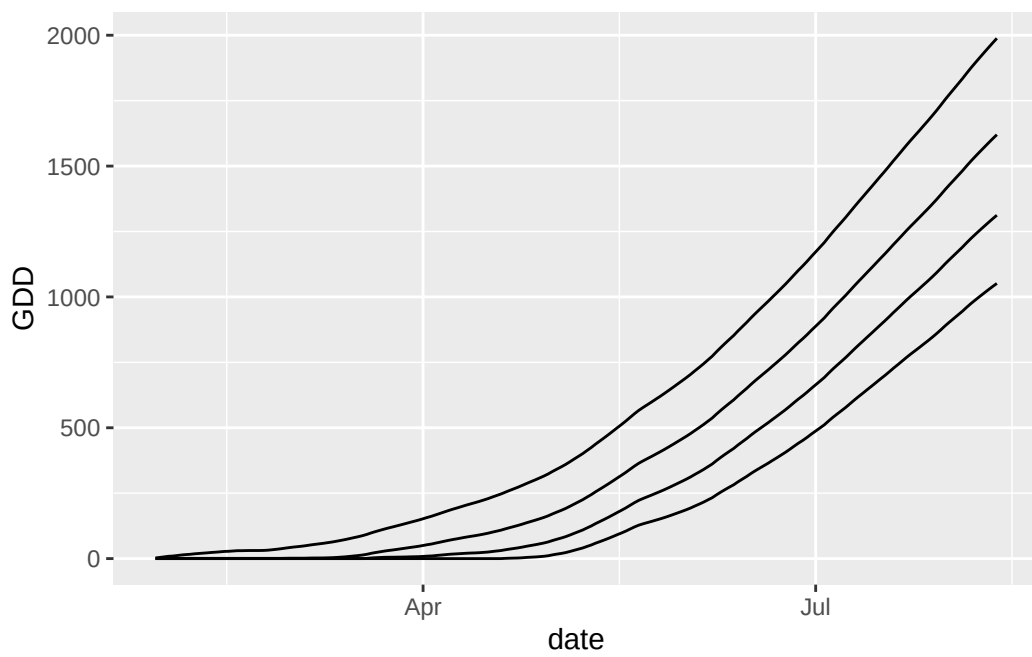
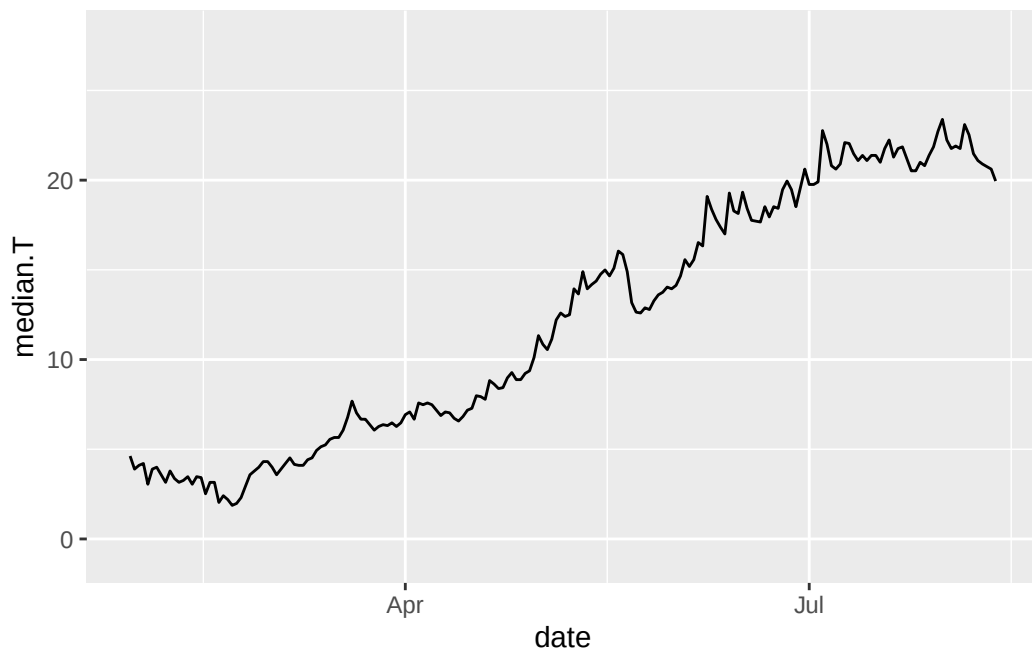
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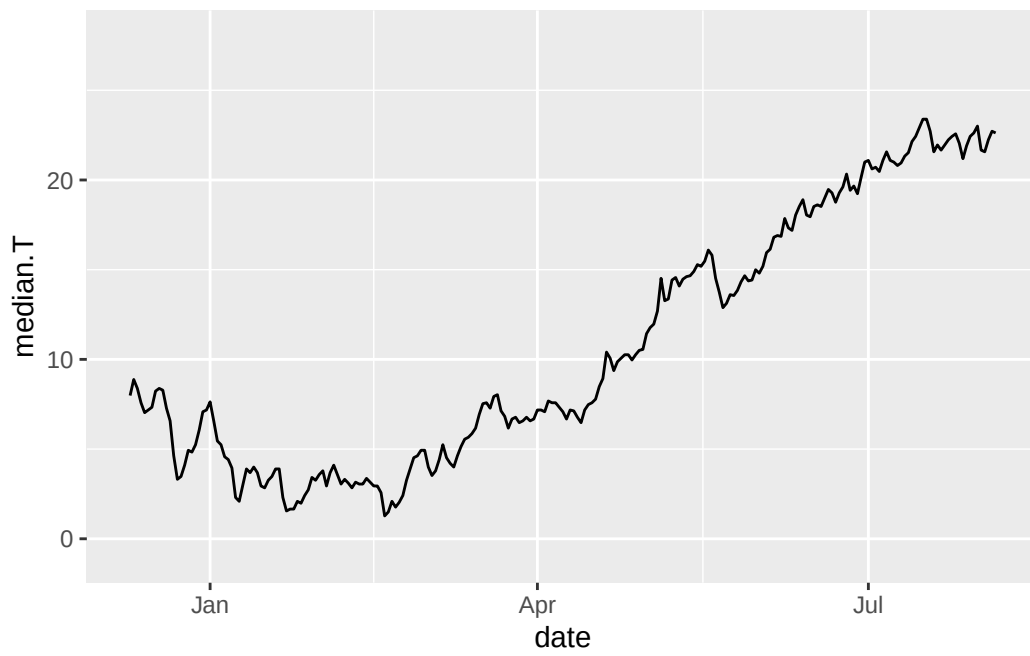
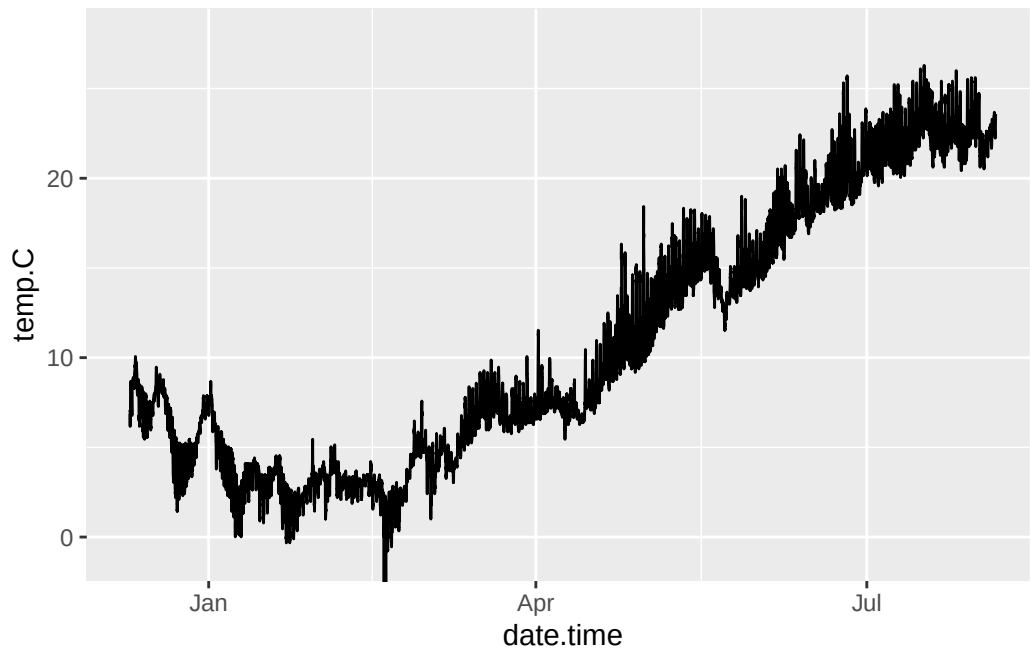


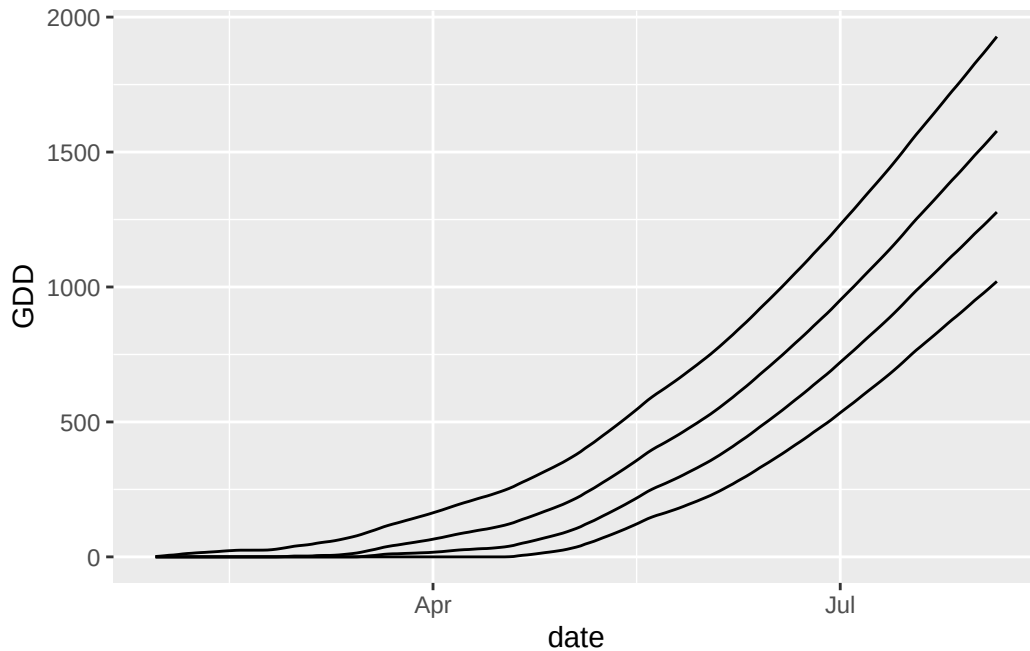
JB*





CL*





Merge HOBO data & Join site coordinates

NASA JPL Multi-scale Ultra-high Resolution (MUR) SST

Extract ERDDAP JPL satellite SST data

jplMURSST41

- NASA JPL
- Acknowledgement: These data were provided by JPL under support by NASA MEaSUREs program.
- Multi-scale Ultra-high Resolution (MUR) SST Analysis fv04.1, Global, 0.01°, 2002-present, Daily
- (<https://coastwatch.pfeg.noaa.gov/erddap/info/jplMURSST41/index.html>)
- Metadata: <https://podaac.jpl.nasa.gov/dataset/MUR-JPL-L4-GLOB-v4.1>

NOAA ACSPO Daily Global 0.02° Gridded Super-collated SST

Extract ERDDAP NOAA ACSPO satellite SST data

noaacwLEOACSPoSSTL3SCDaily - Reanalysis (“Reserach grade”), 2 *month delay*

noaacwLEOACSPoSSTL3SnrtCDaily - Near real time

- Acknowledgement: These data were provided by Group for High Resolution Sea Surface Temperature (GHR SST) and the National Oceanic and Atmospheric Administration (NOAA).
- NOAA/NESDIS/STAR
- Sea-Surface Temperature, NOAA ACSPO Daily Global 0.02° Gridded Super-collated SST and Thermal Fronts Reanalysis, 2012-present, Daily (L3S-LEO degrees C)
- (<https://coastwatch.noaa.gov/erddap/info/noaacwLEOACSPoSSTL3SCDaily/index.html>)

NOAA Coral Reef Watch Daily Global 5km Satellite SST (CoralTemp)

Extract ERDDAP NOAA Coral Reef Watch Daily Global 5km Satellite SST (CoralTemp)

noaacrwsstDaily

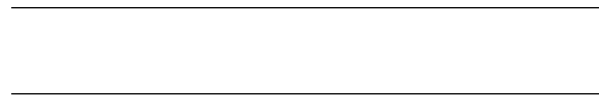
- Acknowledgement: NOAA Coral Reef Watch Program
- NOAA/NESDIS/STAR Coral Reef Watch Program
- Sea Surface Temperature, NOAA Coral Reef Watch Daily Global 5km Satellite SST (CoralTemp), 1985-present, Daily
- (<https://coastwatch.noaa.gov/erddap/info/noaacrwsstDaily/index.html>)

NOAA Geo-polar Blended Analysis Night Only, GHR SST, 5km SST

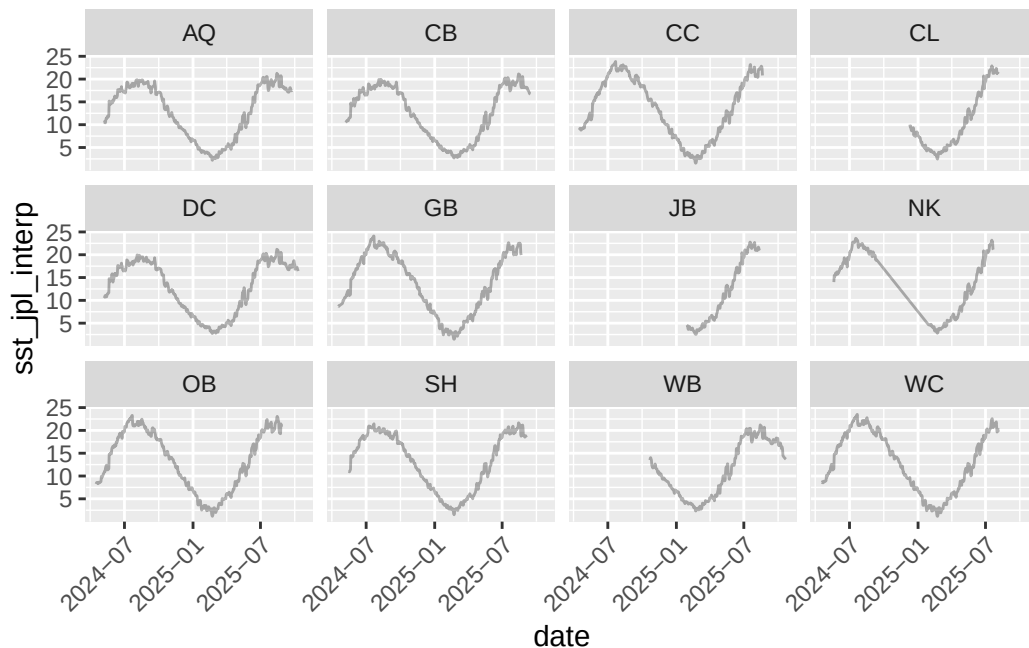
Extract ERDDAP NOAA Geo-polar Blended Analysis Night Only, GHR SST, 5km SST

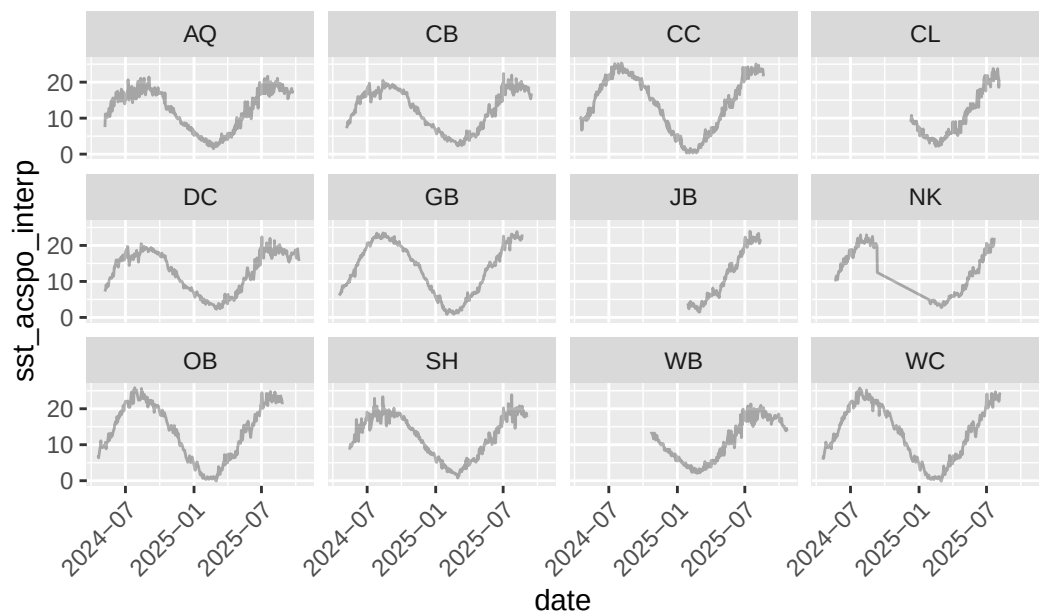
noaacwBLENDEDCsstDaily

- Acknowledgement: NOAA/NESDIS
- Sea-Surface Temperature, NOAA Geo-polar Blended Analysis Night Only, GHR SST, Near Real-Time, Global 5km, 2002-Present, Daily (degree C)
- The Geo-Polar Blended Sea Surface Temperature (SST) Analysis combines multi-satellite retrievals of sea surface temperature into a single analysis of SST. This analysis includes only nighttime data.
- (<https://coastwatch.noaa.gov/erddap/info/noaacwBLENDEDCsstDaily/index.html>)

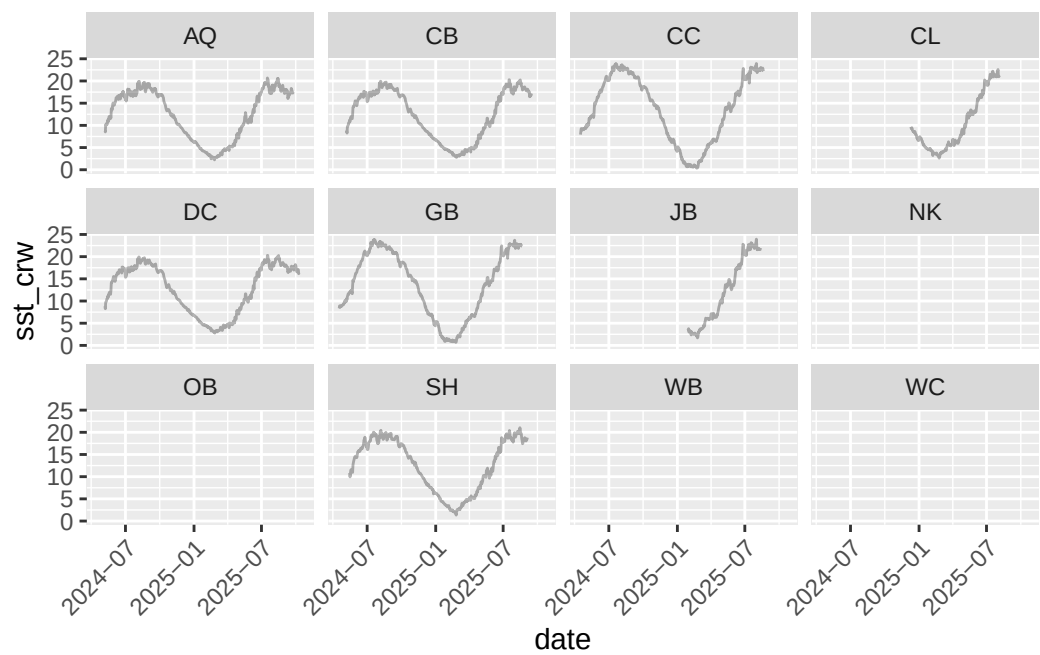


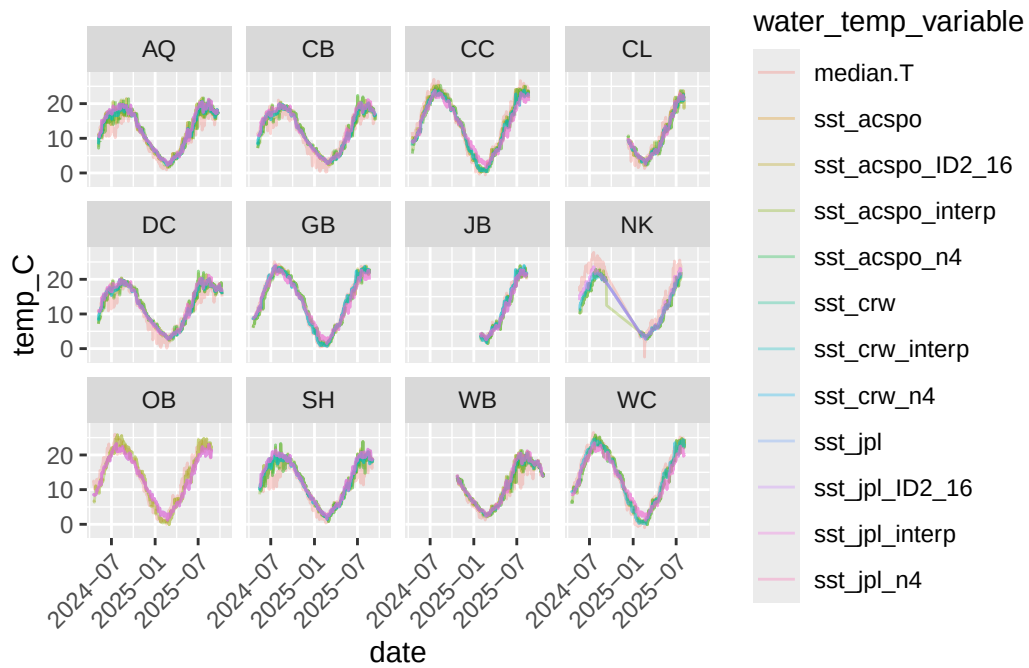
Build complete dataframe with all temp data for analysis





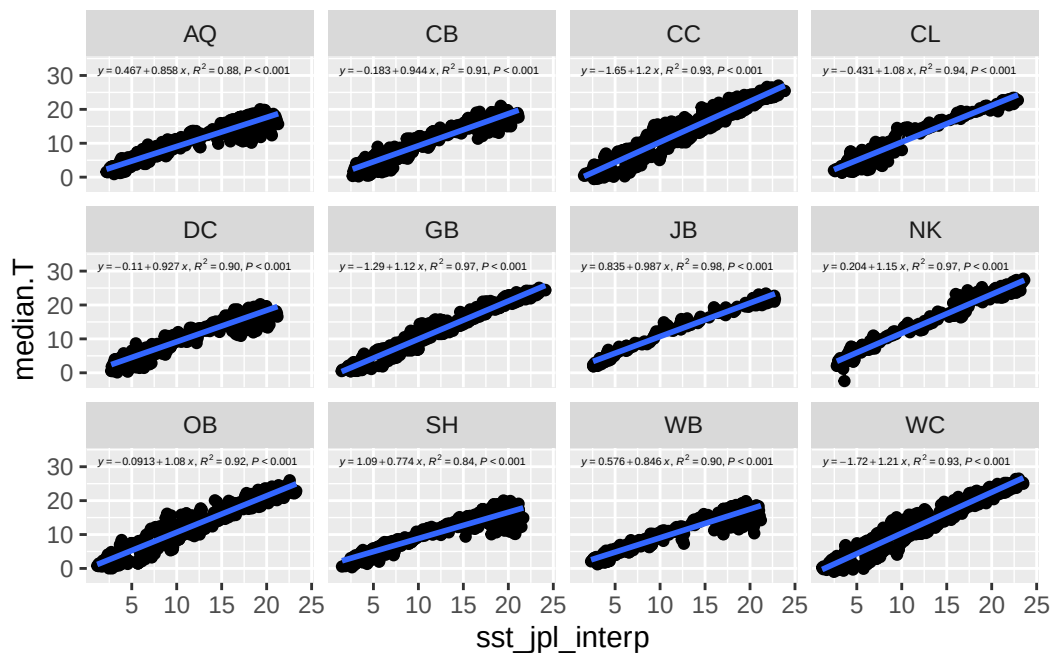
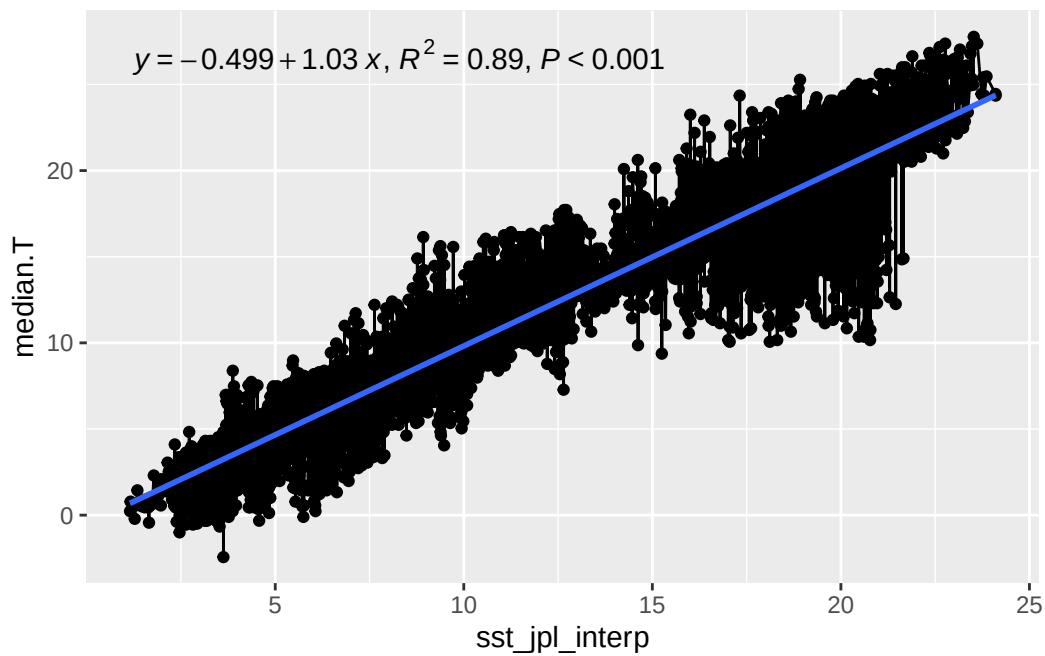
Note: Linear interpolation used to infill missing values



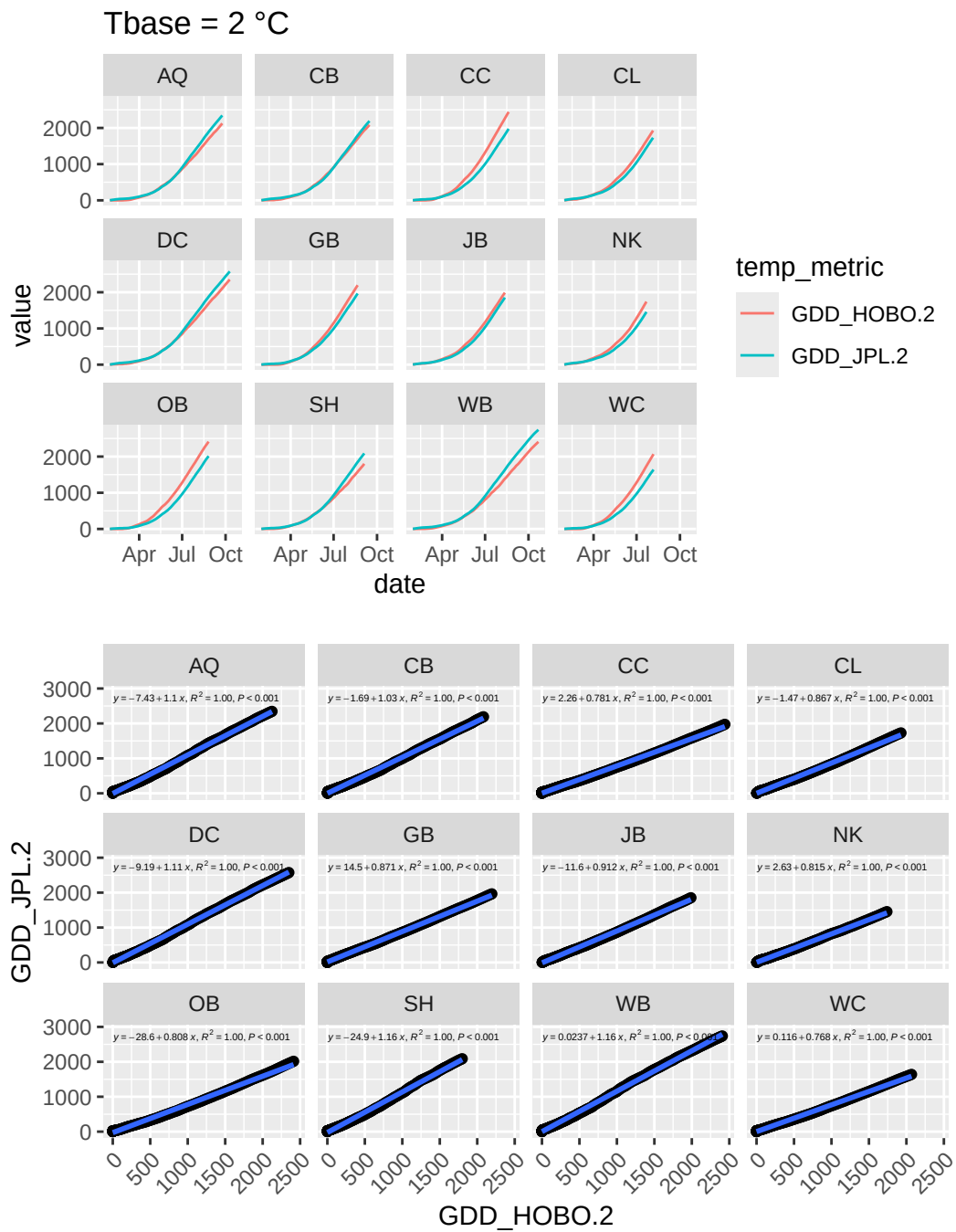


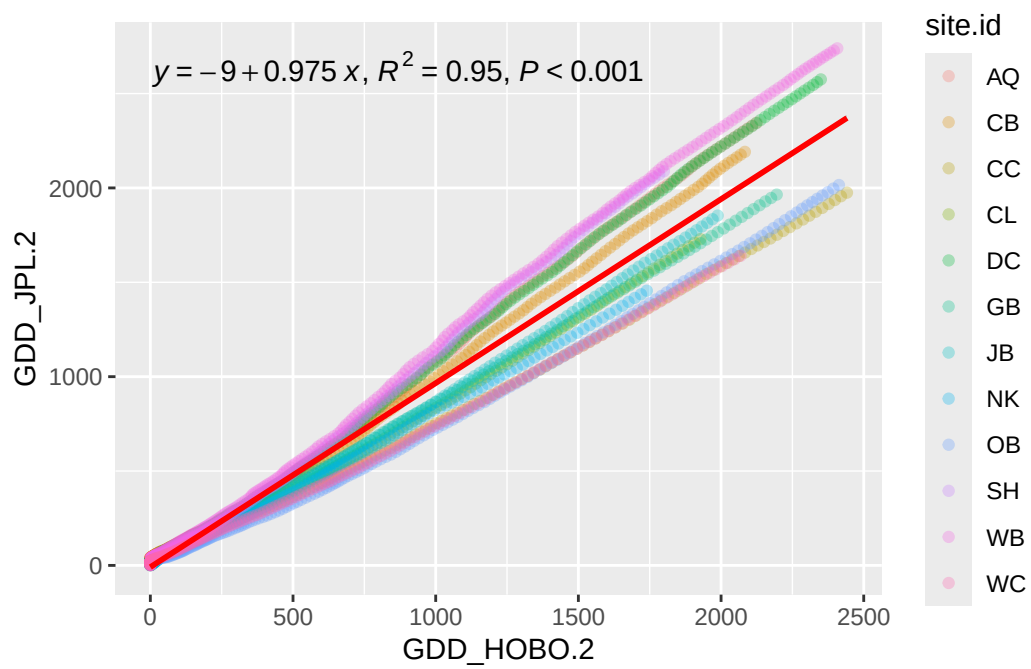
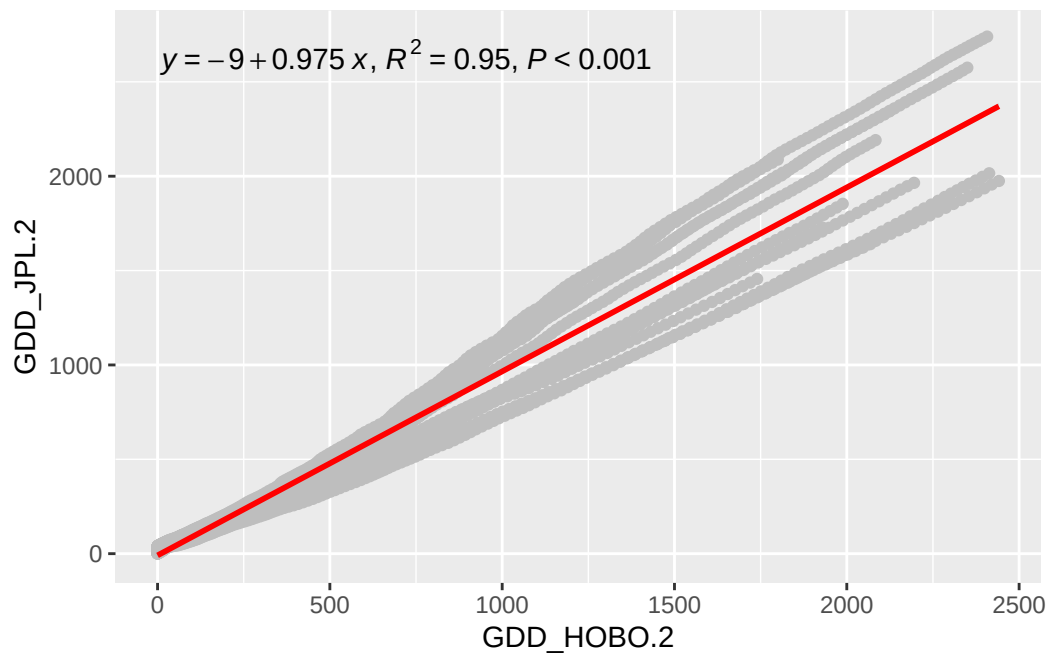
Build GDD dataframes for each satellite dataset

Compare HOBO and NASA JPL satellite data

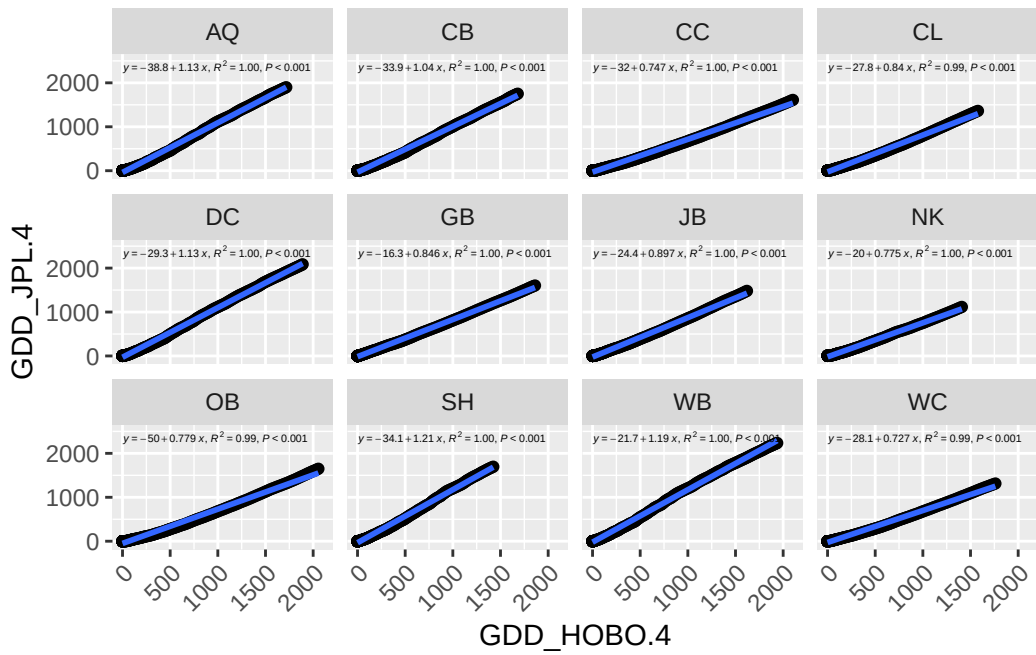
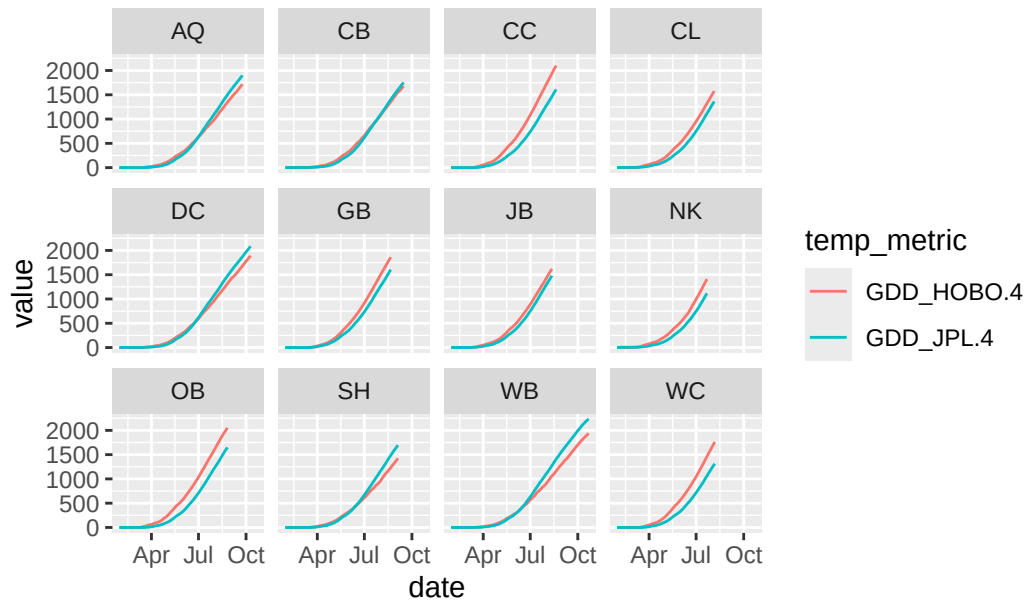


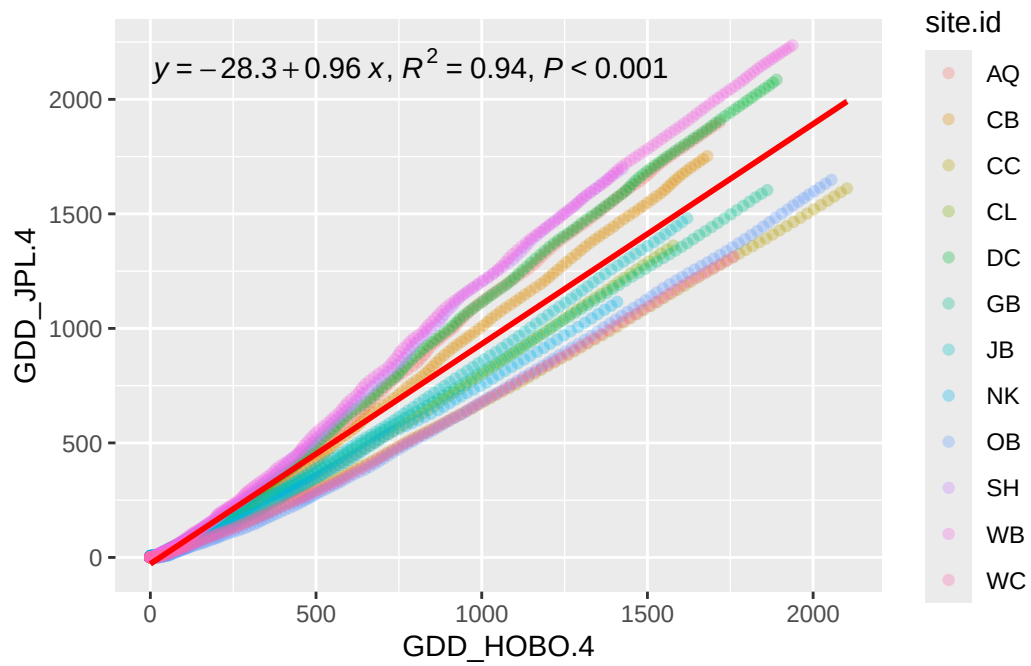
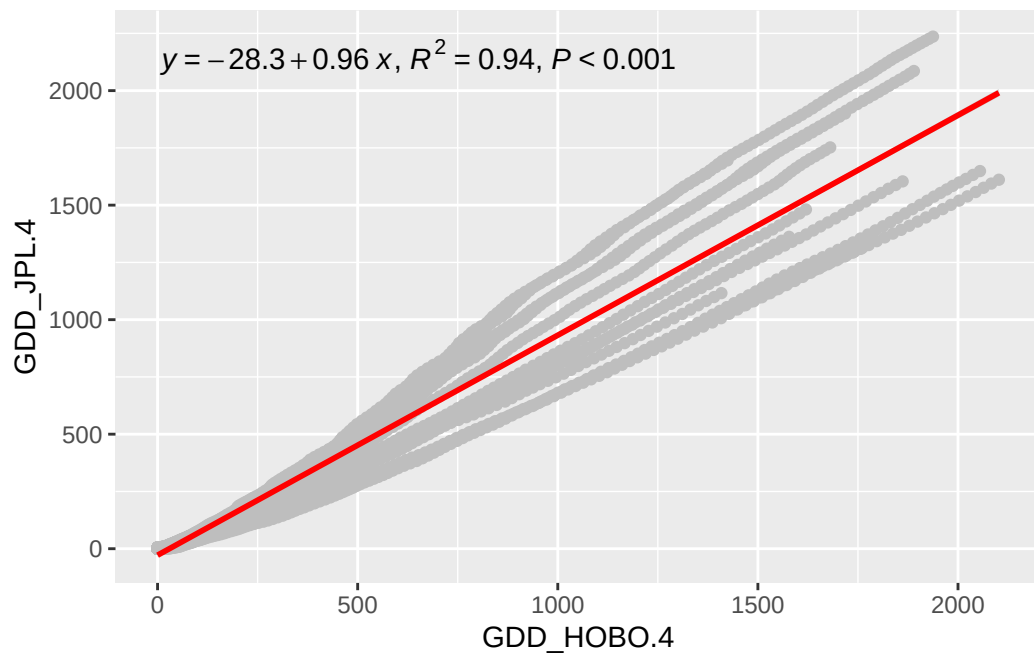
Compare Growing Degree Days (GDD) - NASA JPL



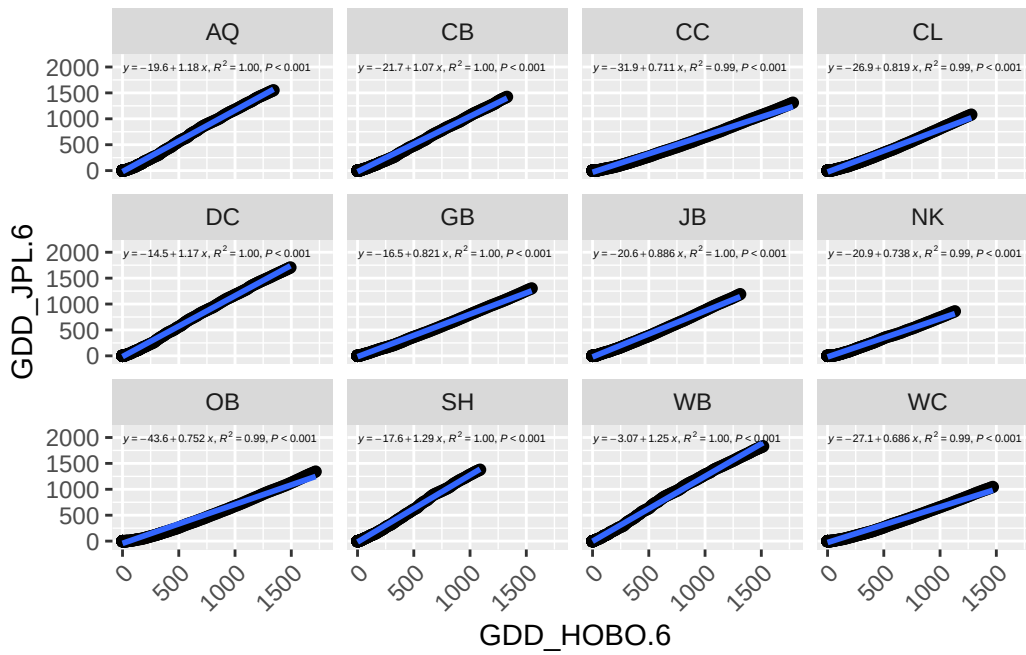
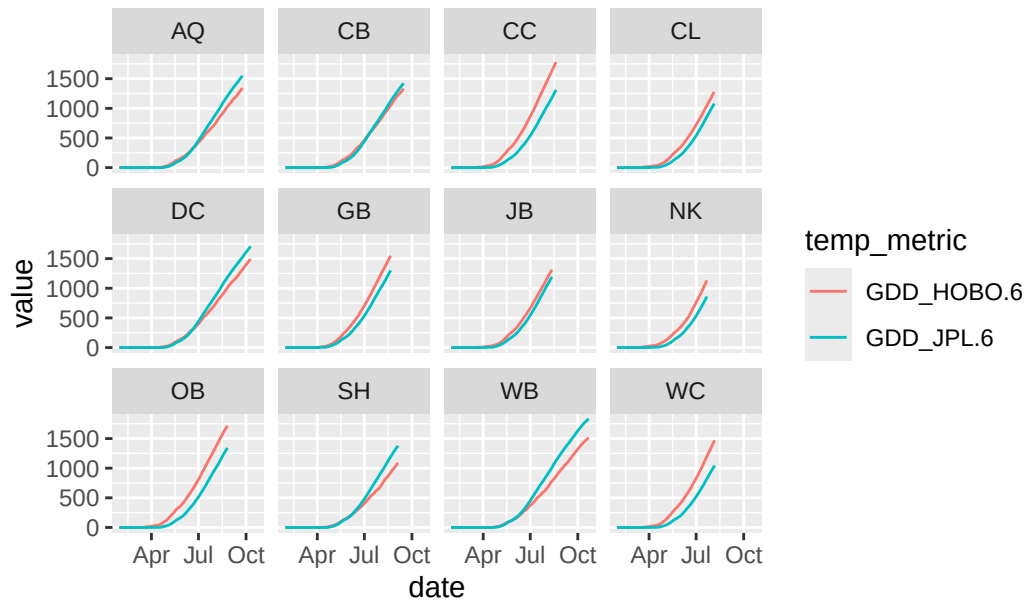


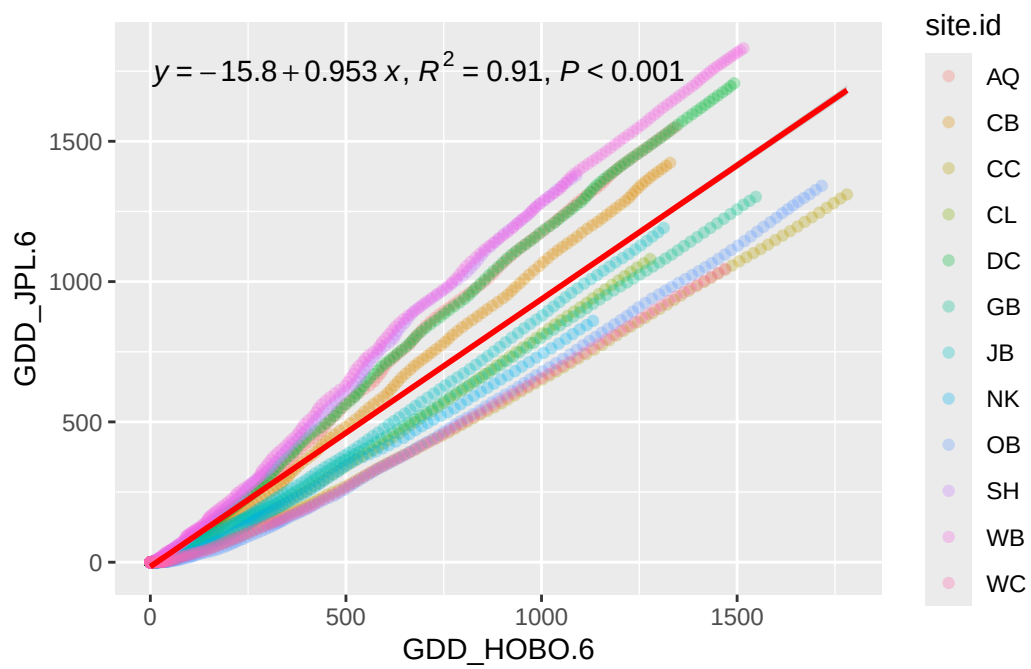
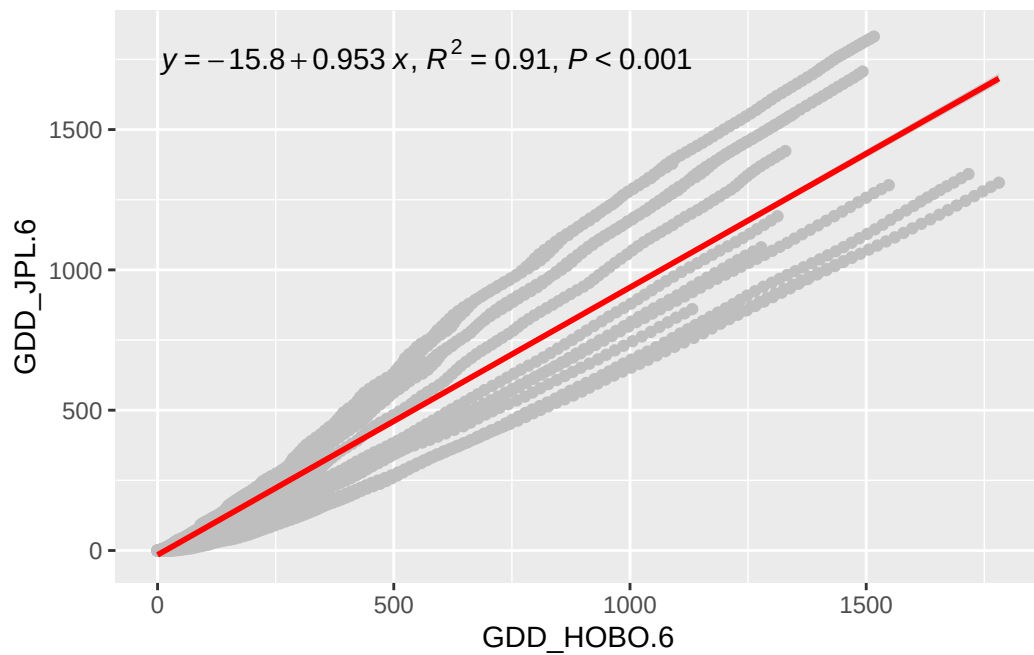
Tbase = 4 °C

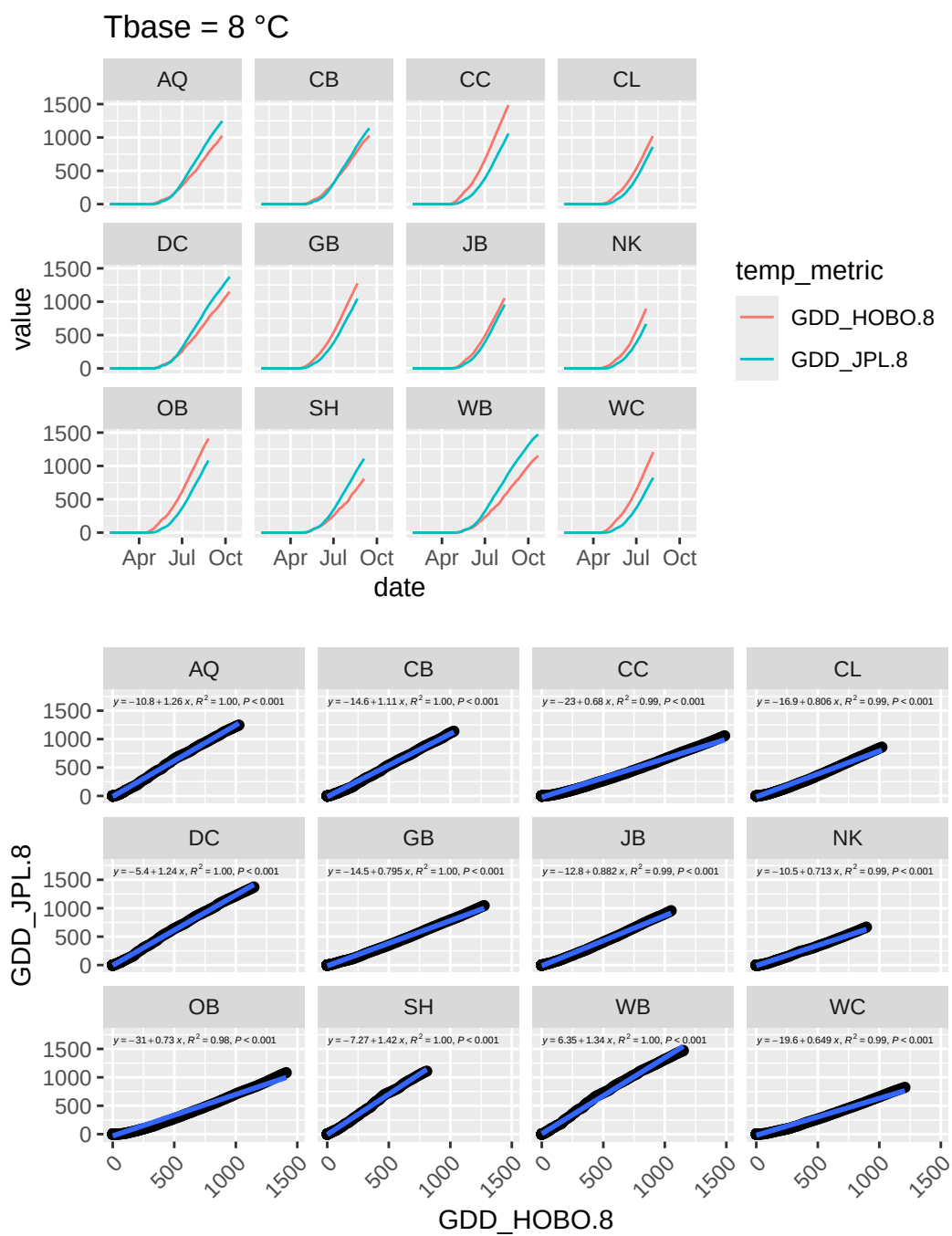


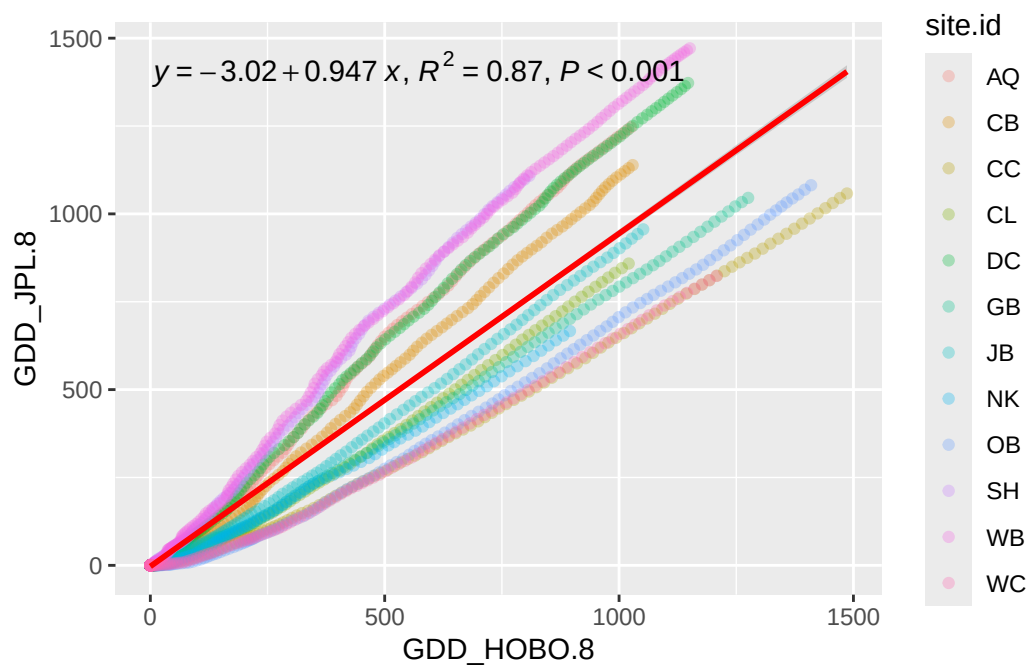
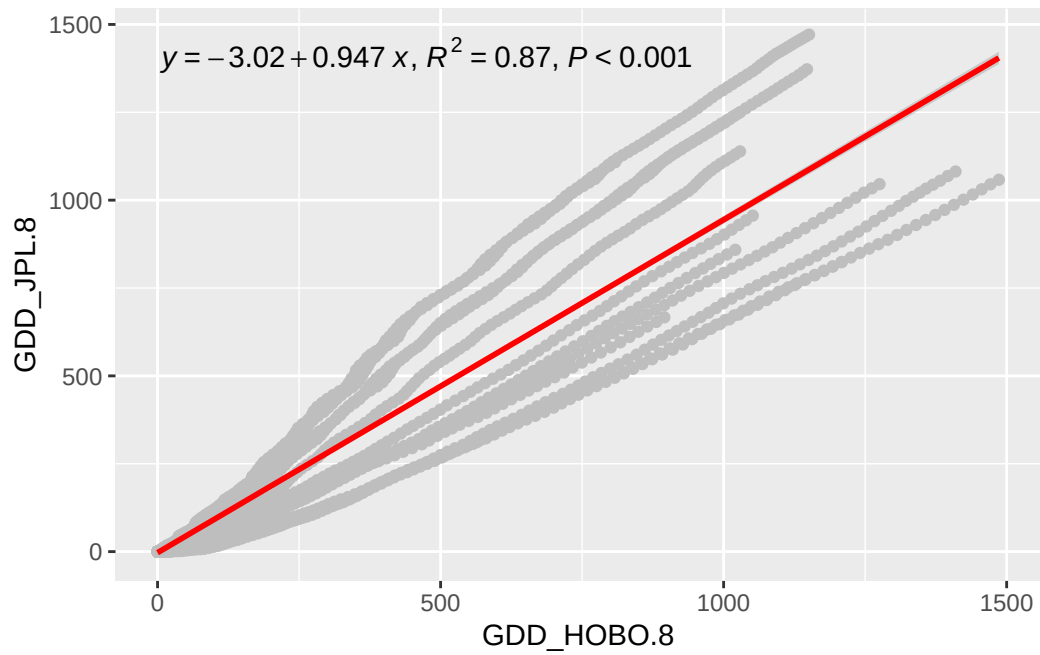


Tbase = 6 °C

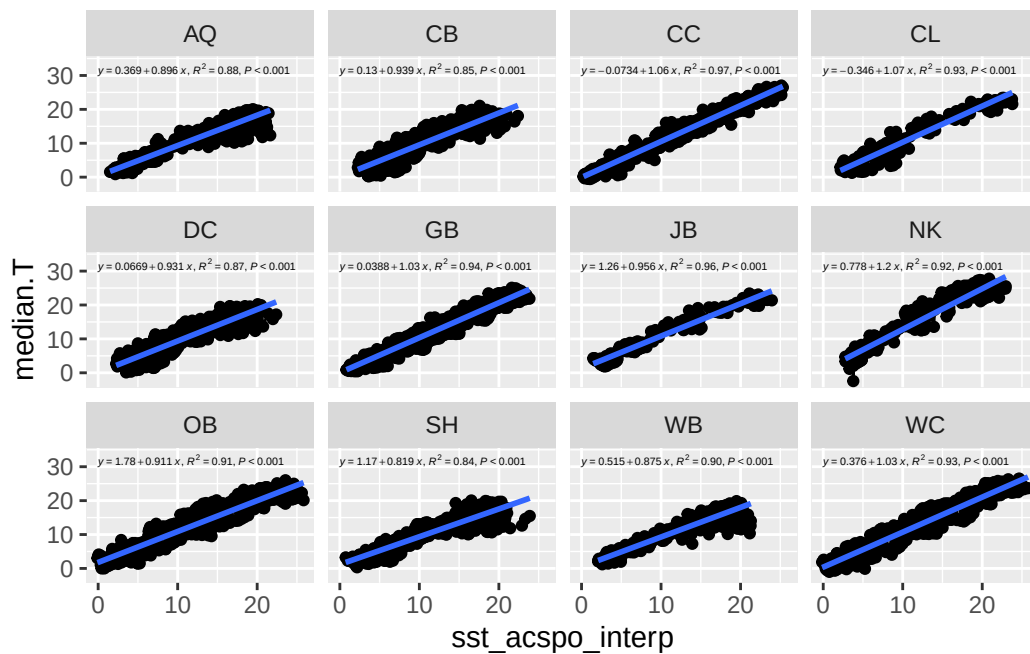
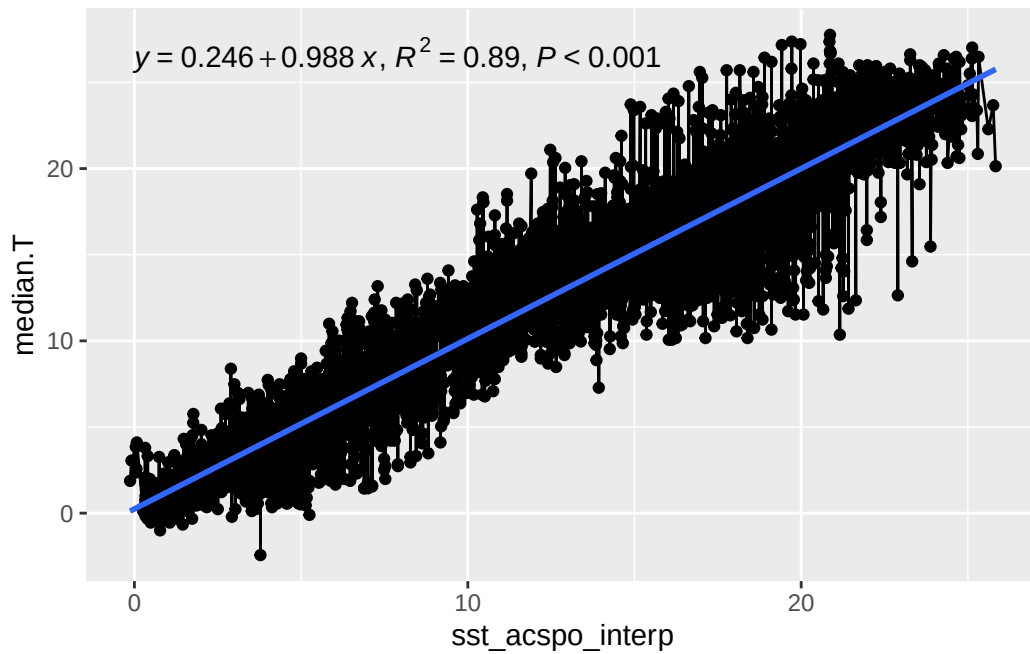




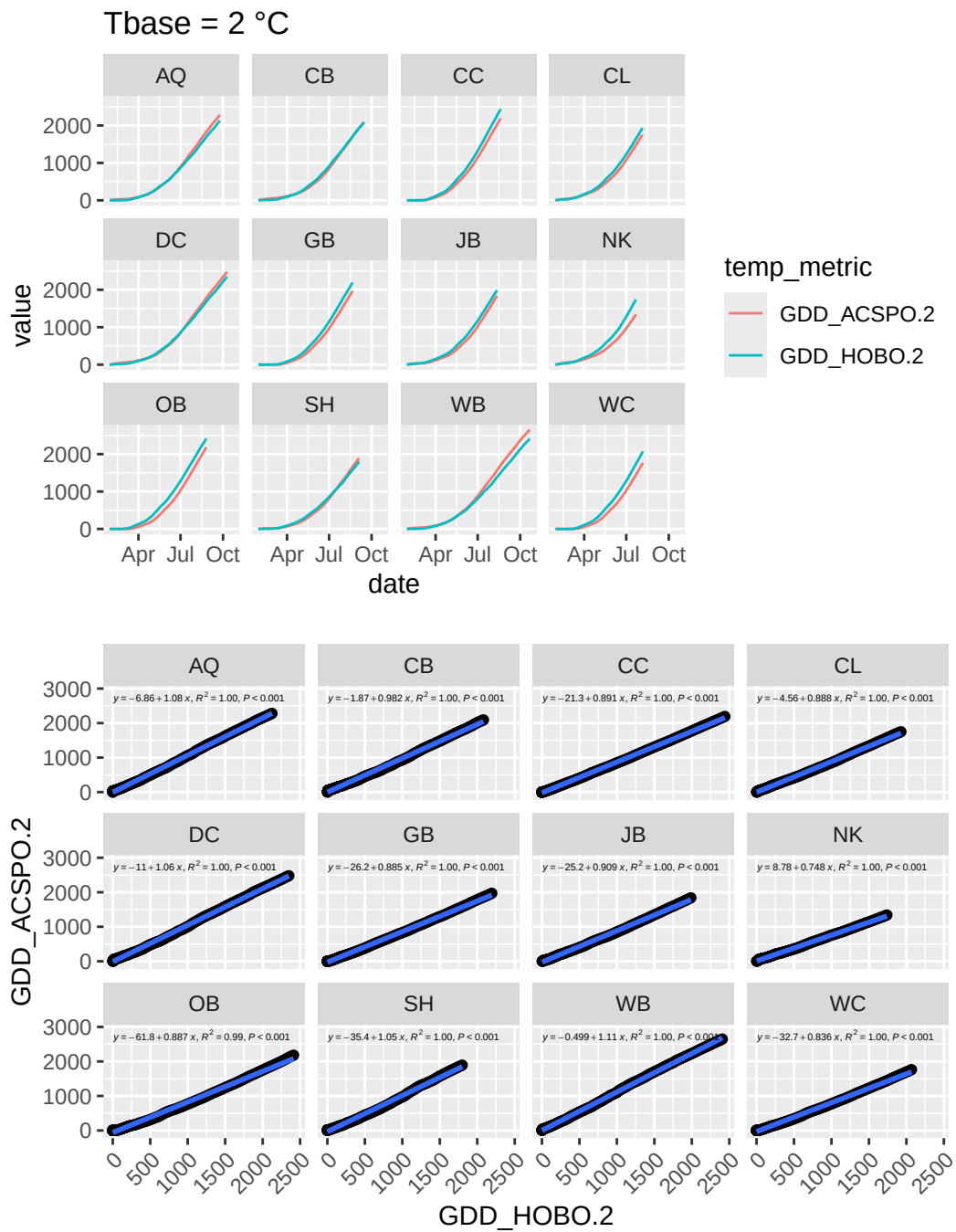


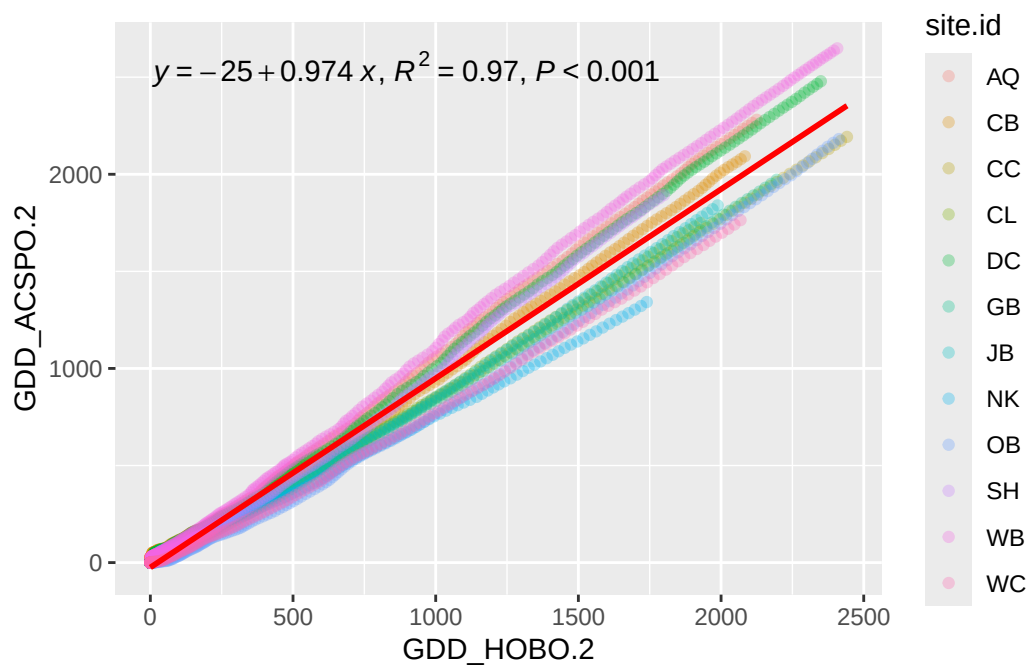
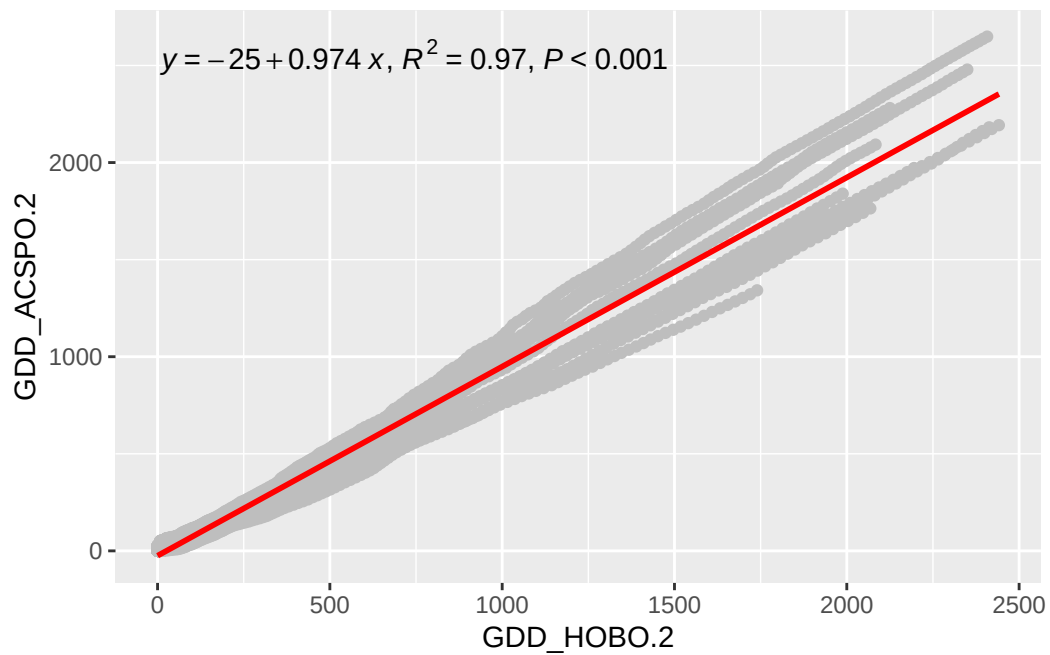


Compare HOBO and NASA ACSPO satellite data

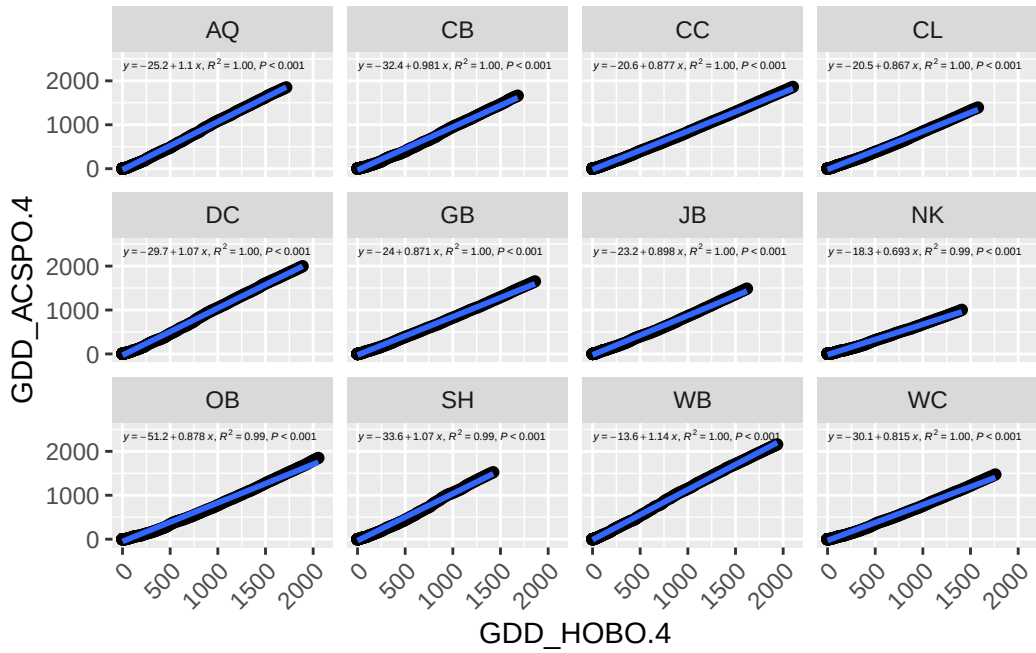
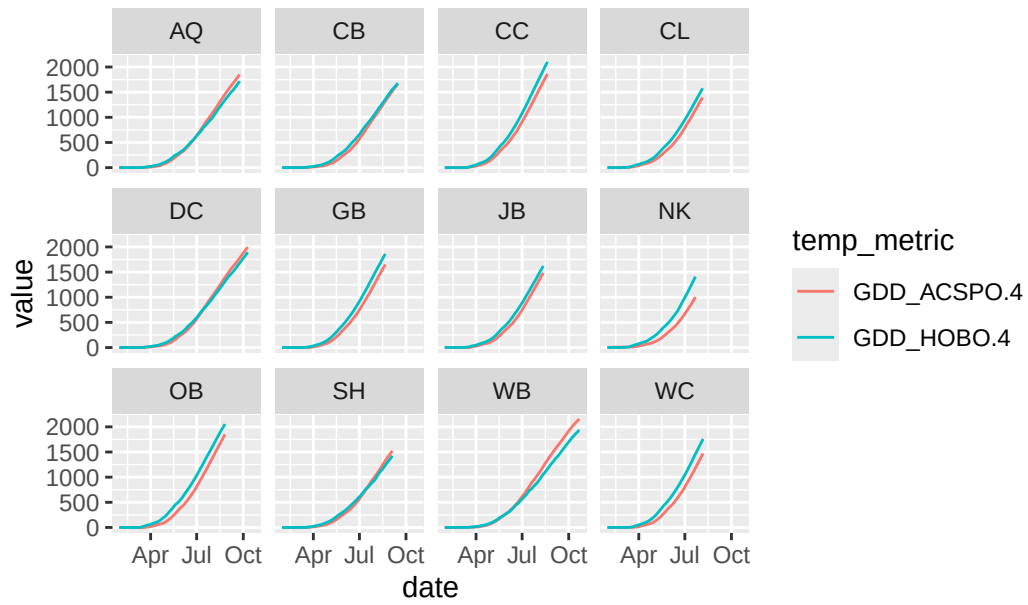


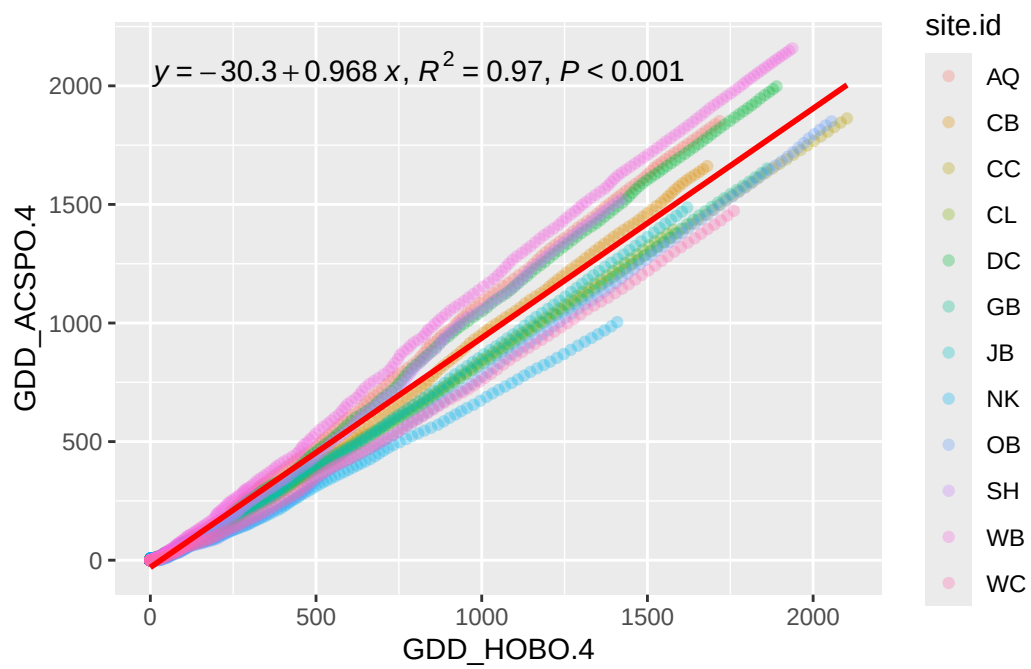
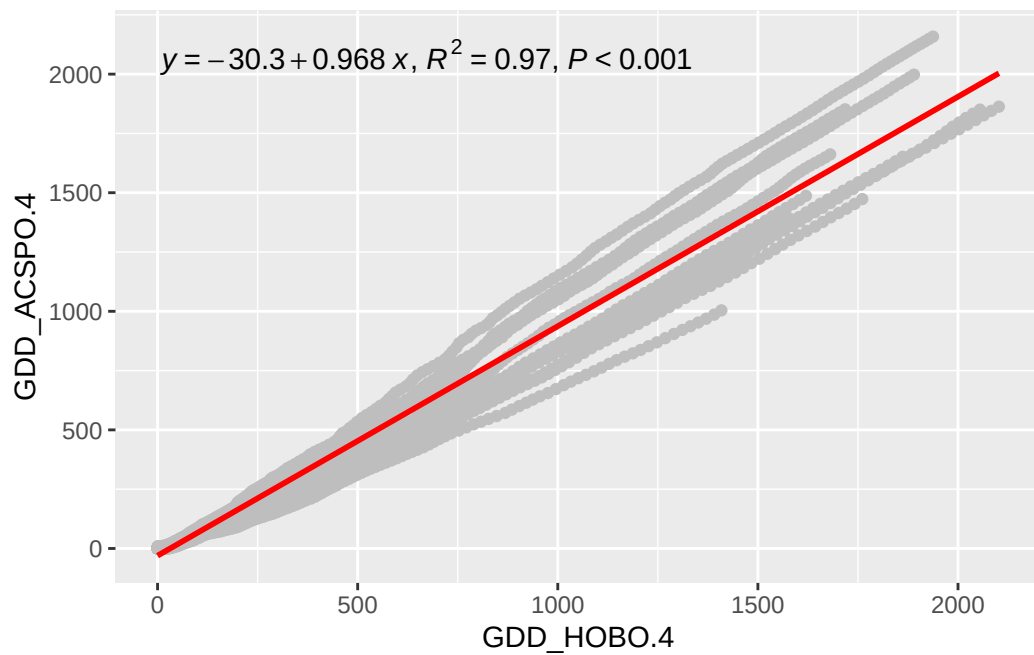
Compare Growing Degree Days (GDD) - NASA ACSPO



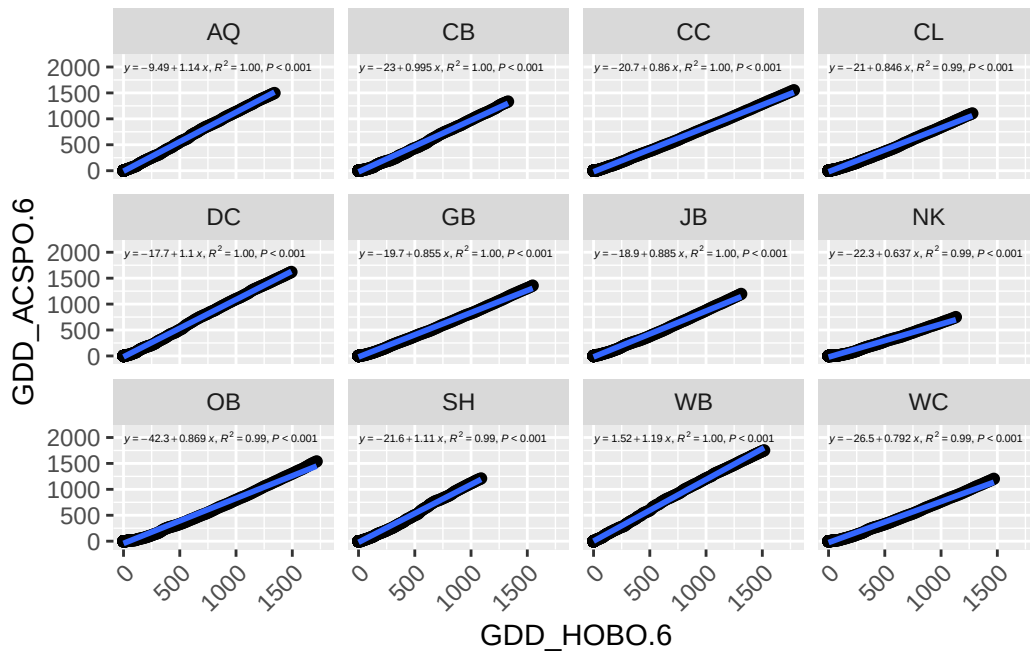
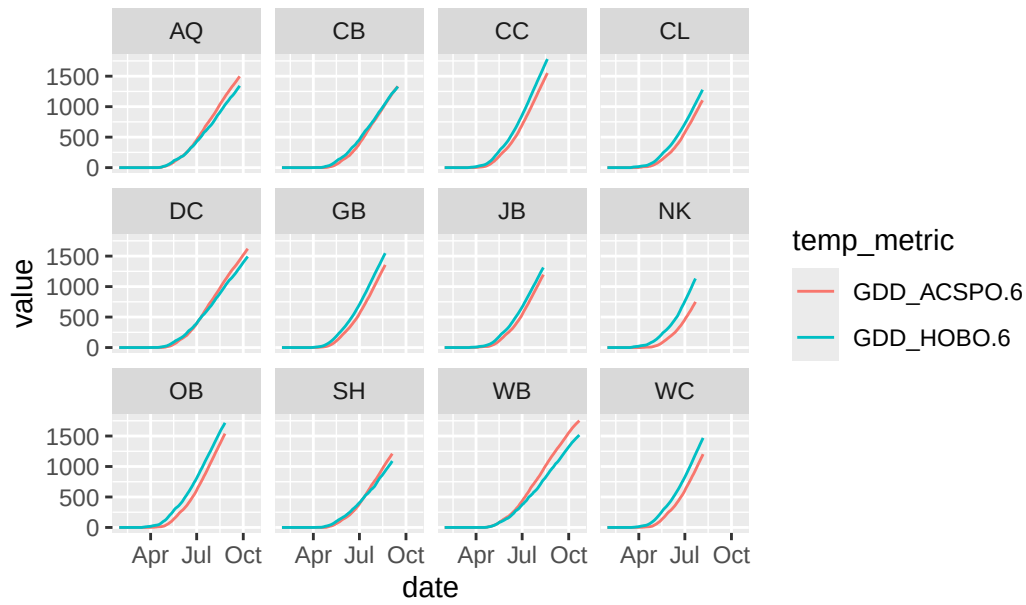


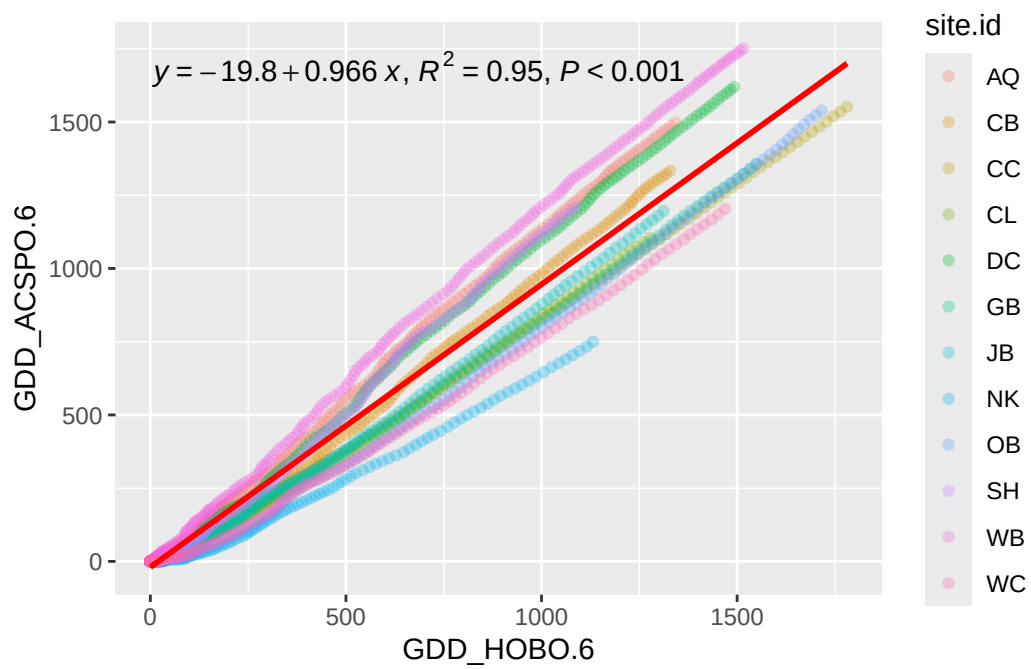
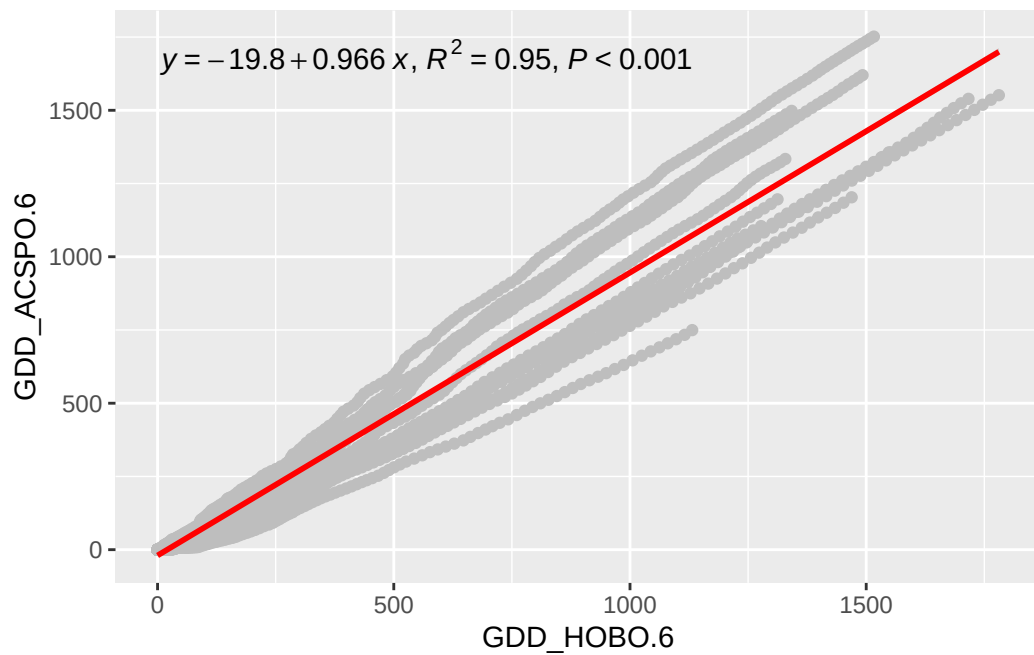
Tbase = 4 °C

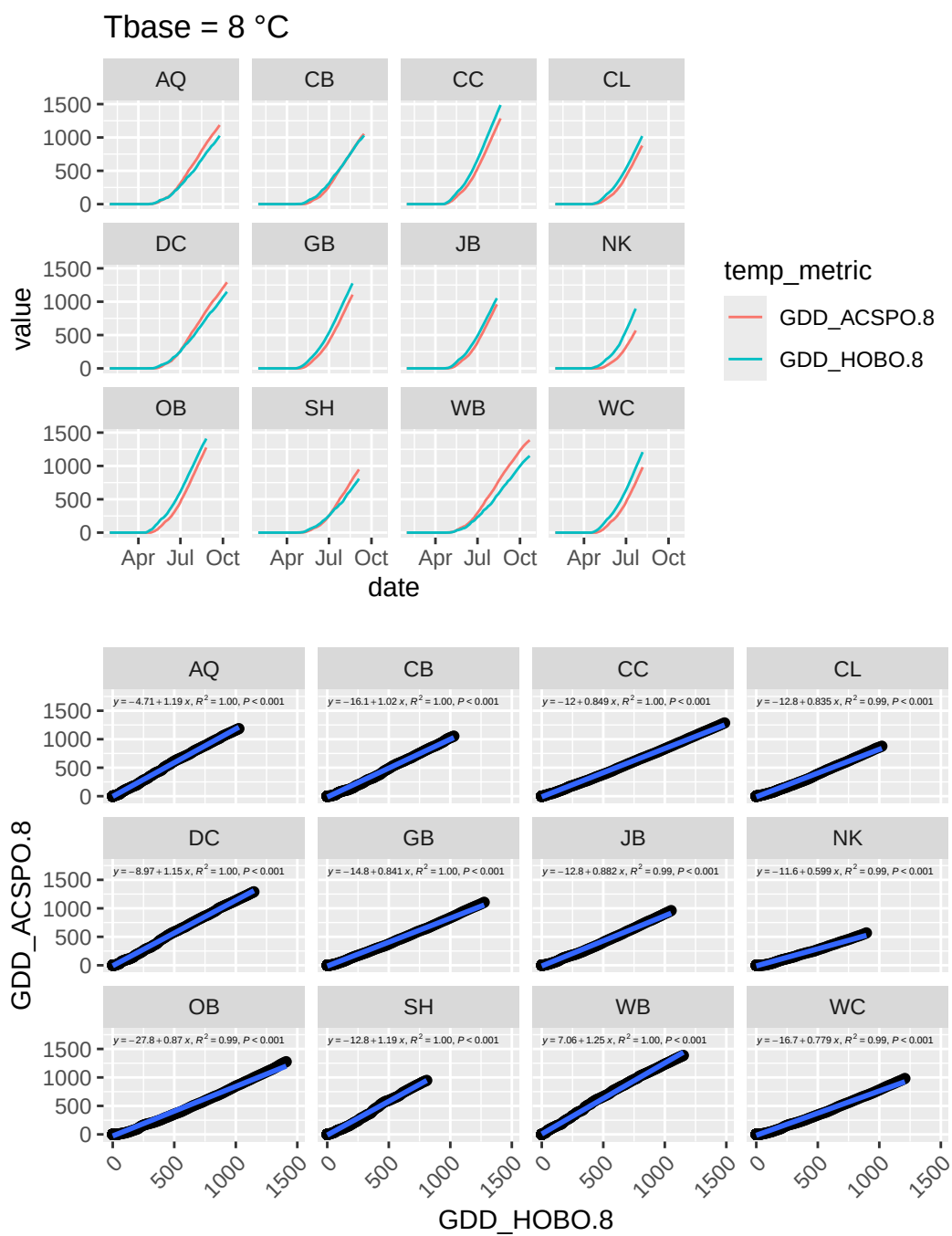


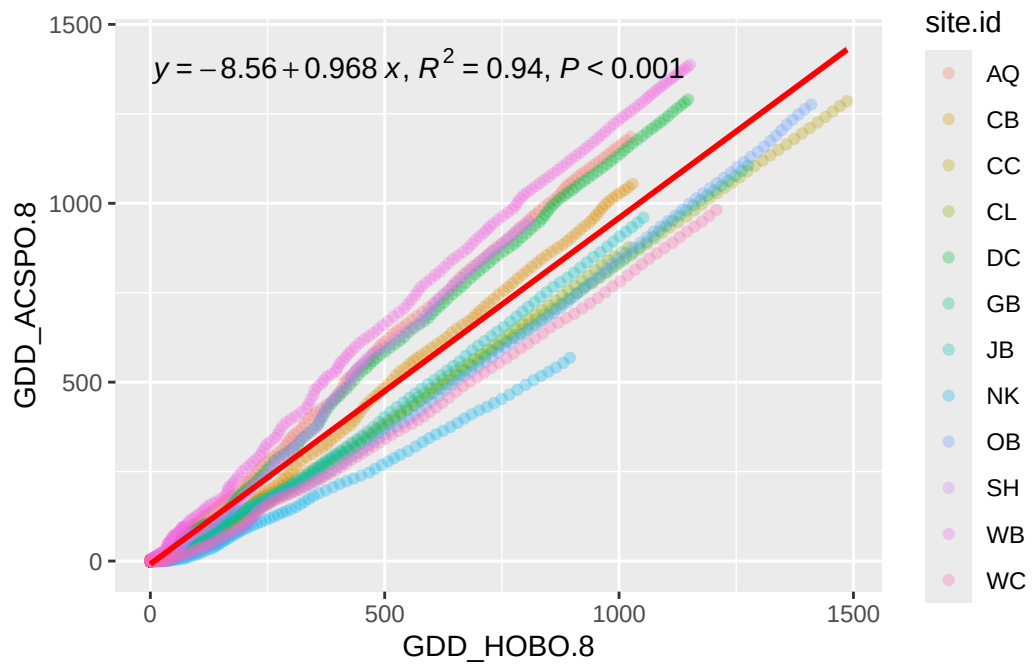
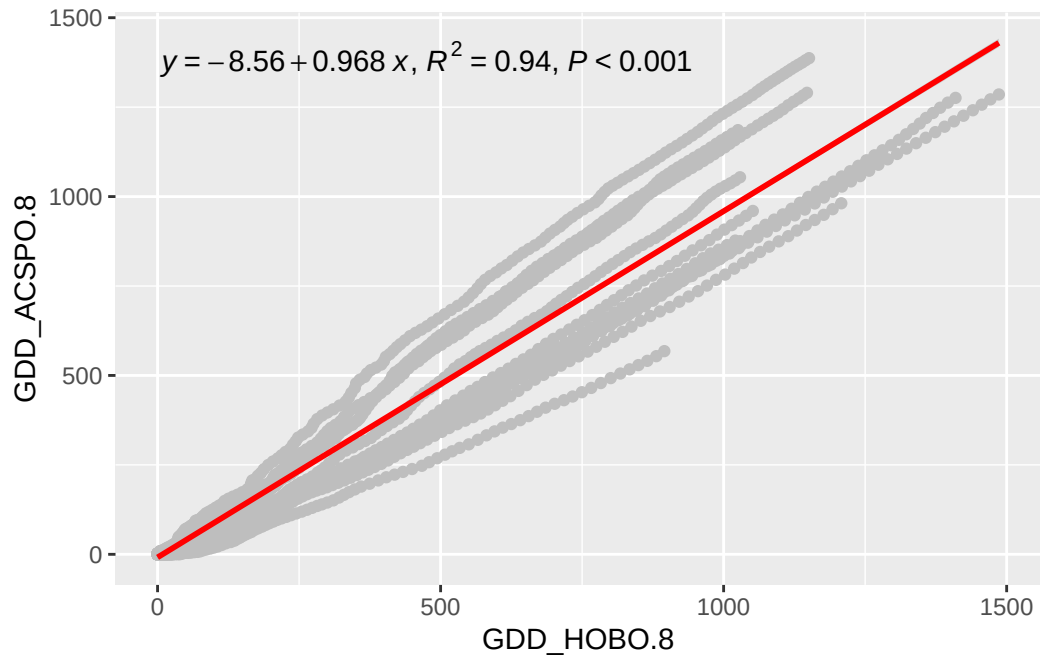


Tbase = 6 °C

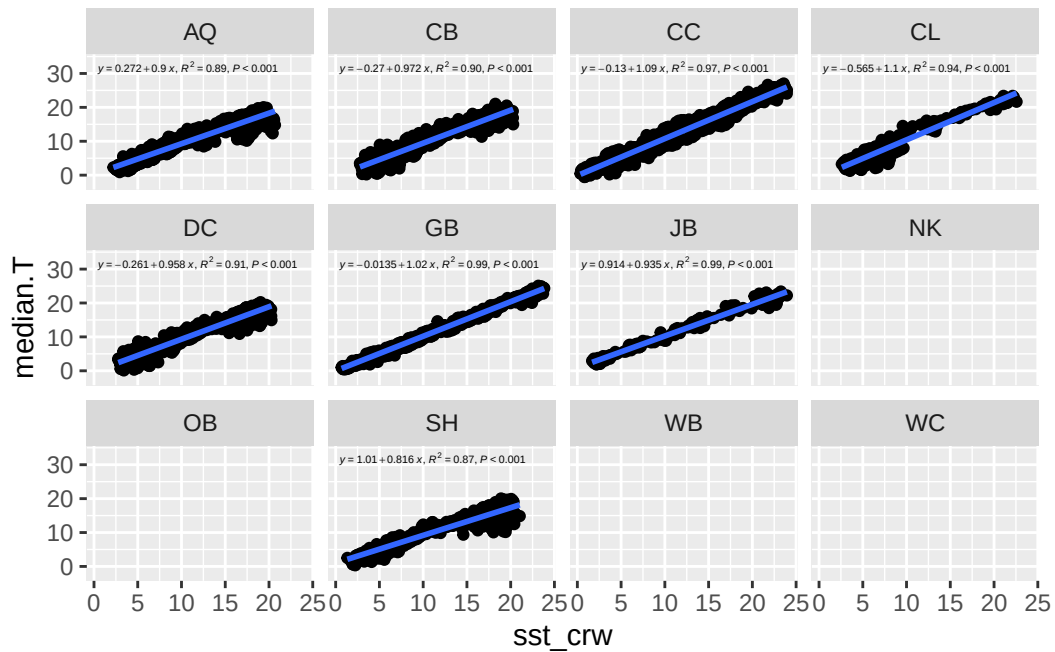
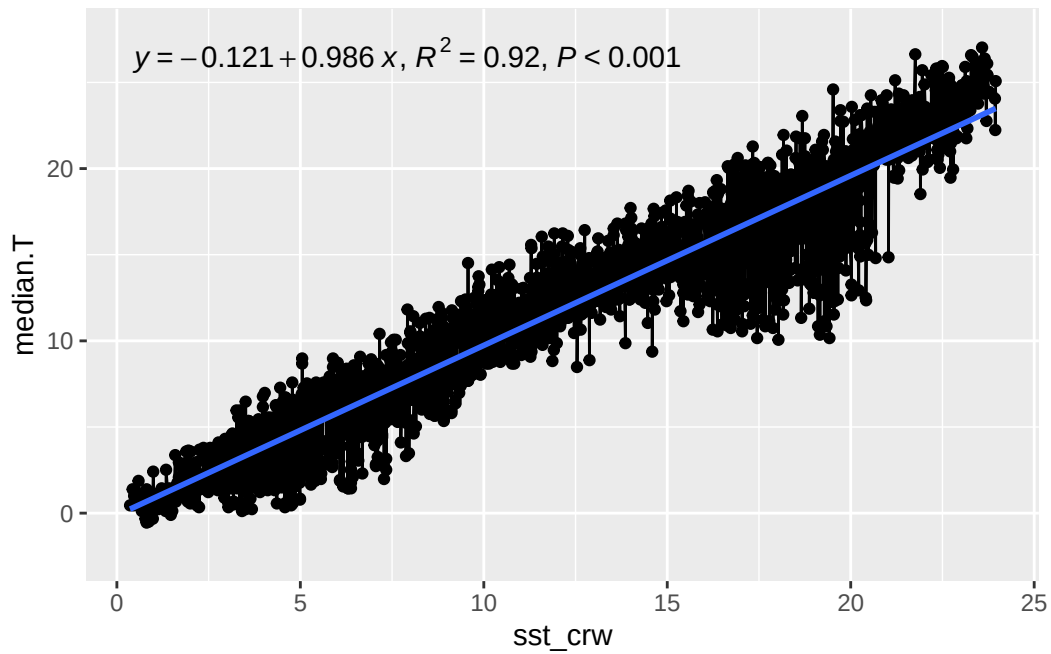




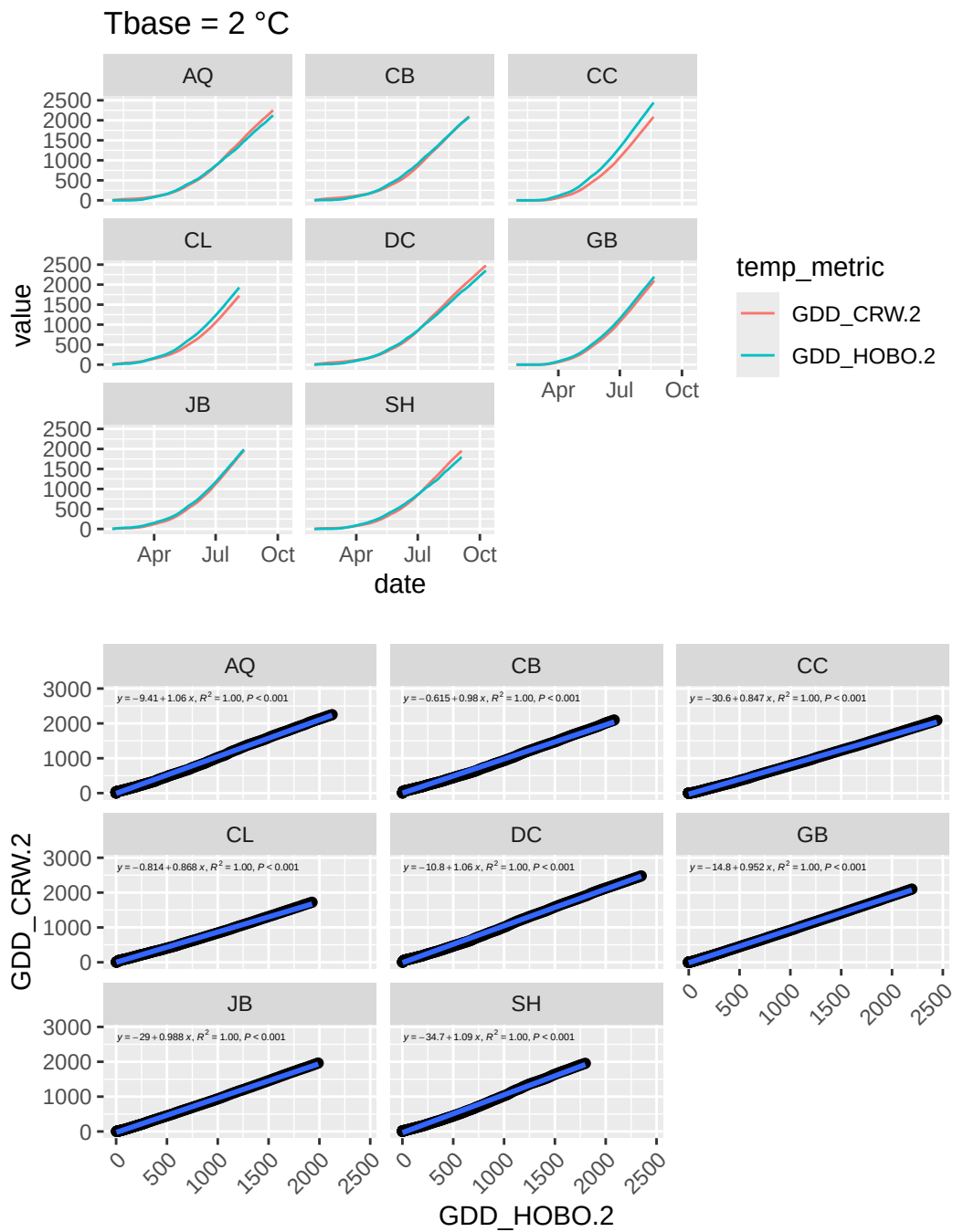


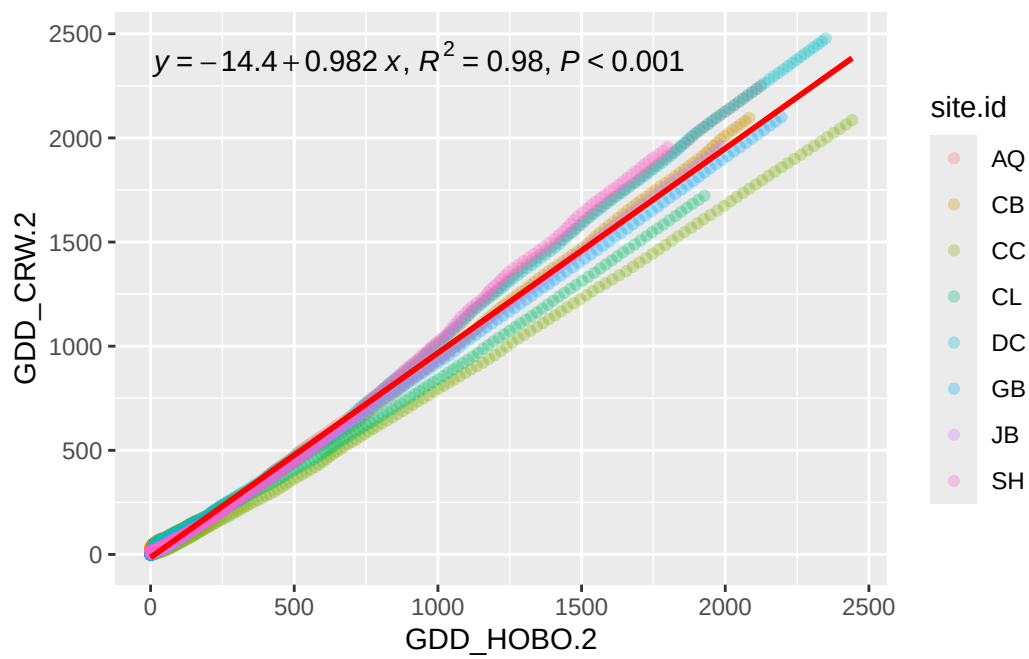
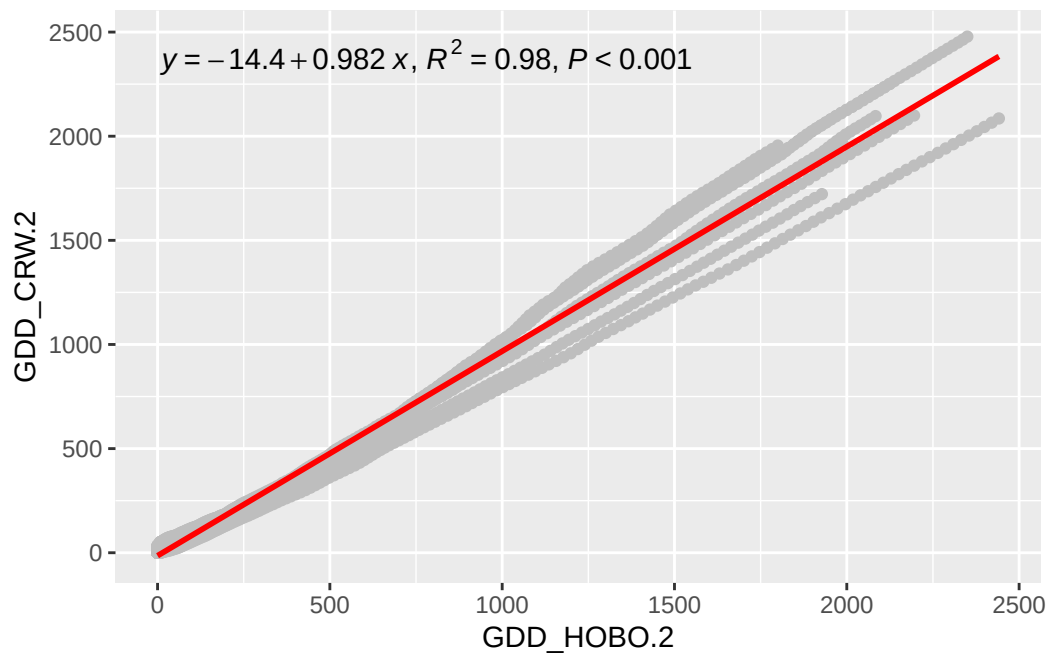


Compare HOBO and NASA CRW satellite data

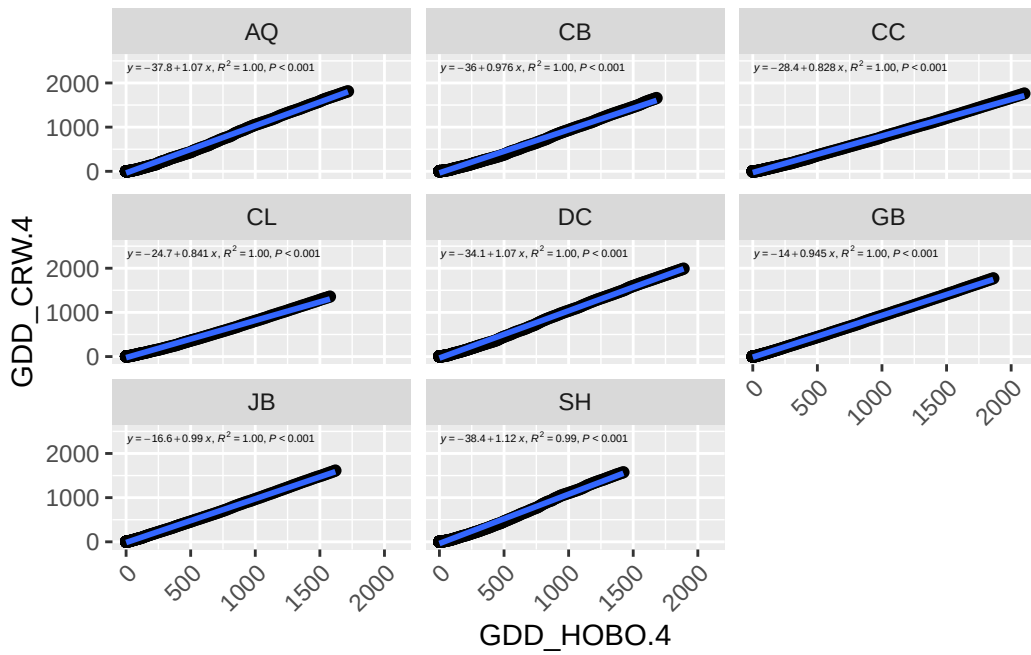
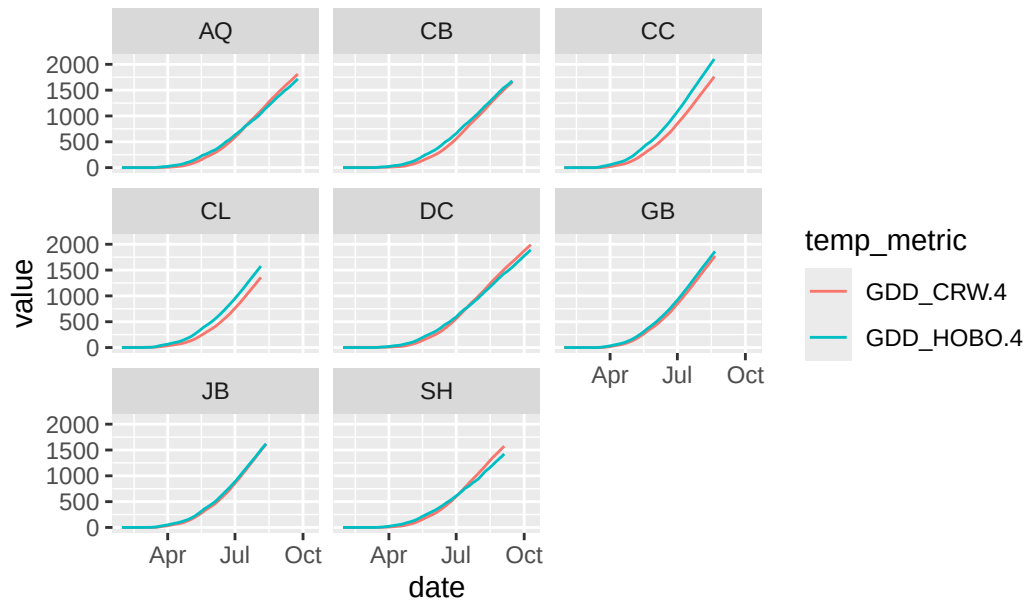


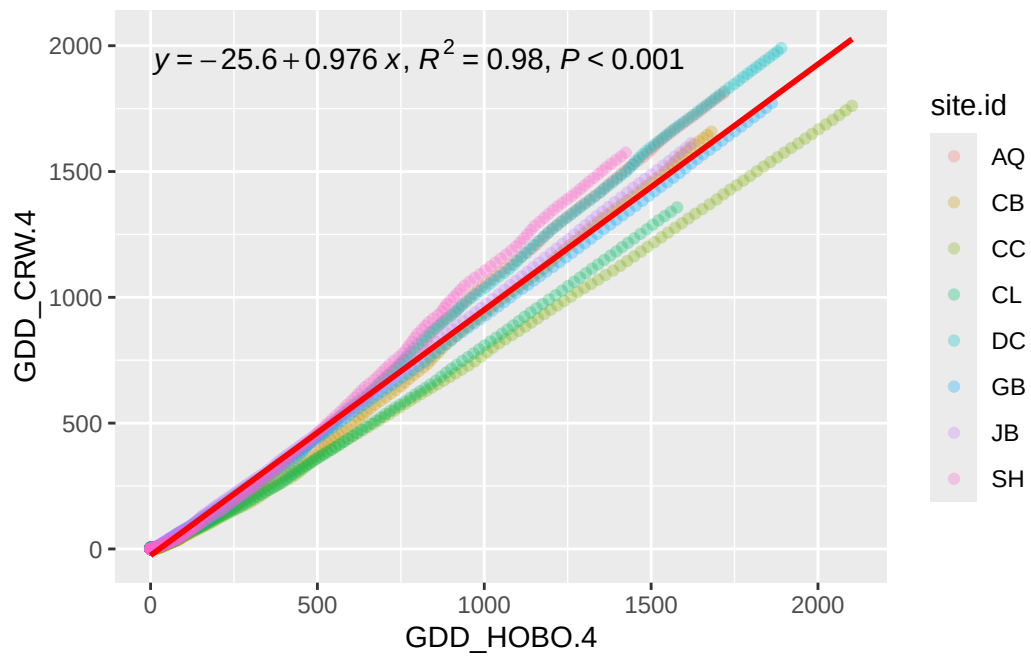
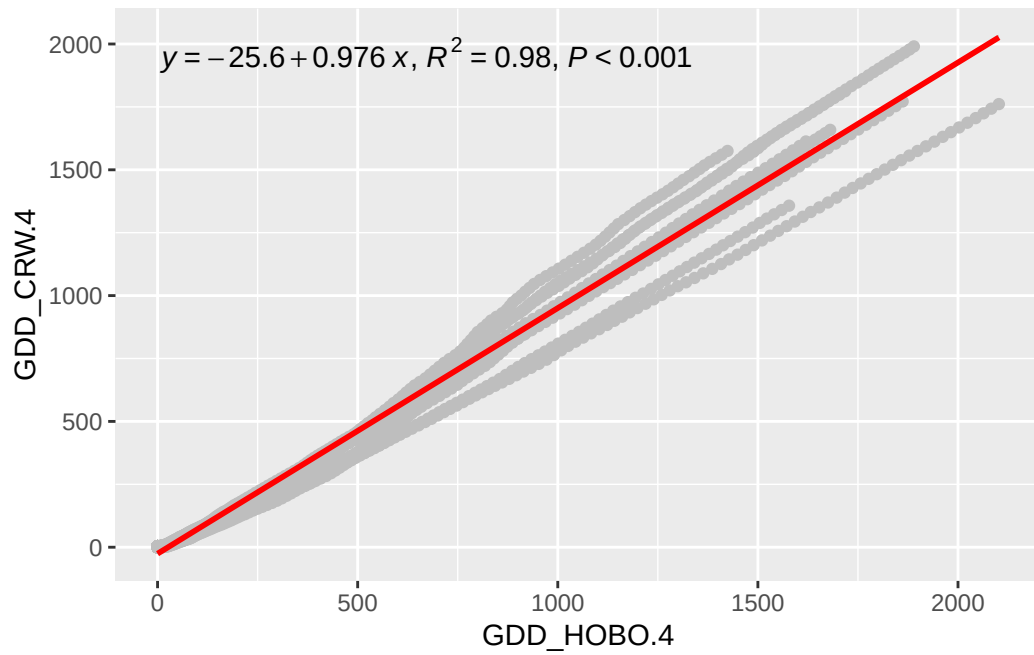
Compare Growing Degree Days (GDD) - NASA CRW



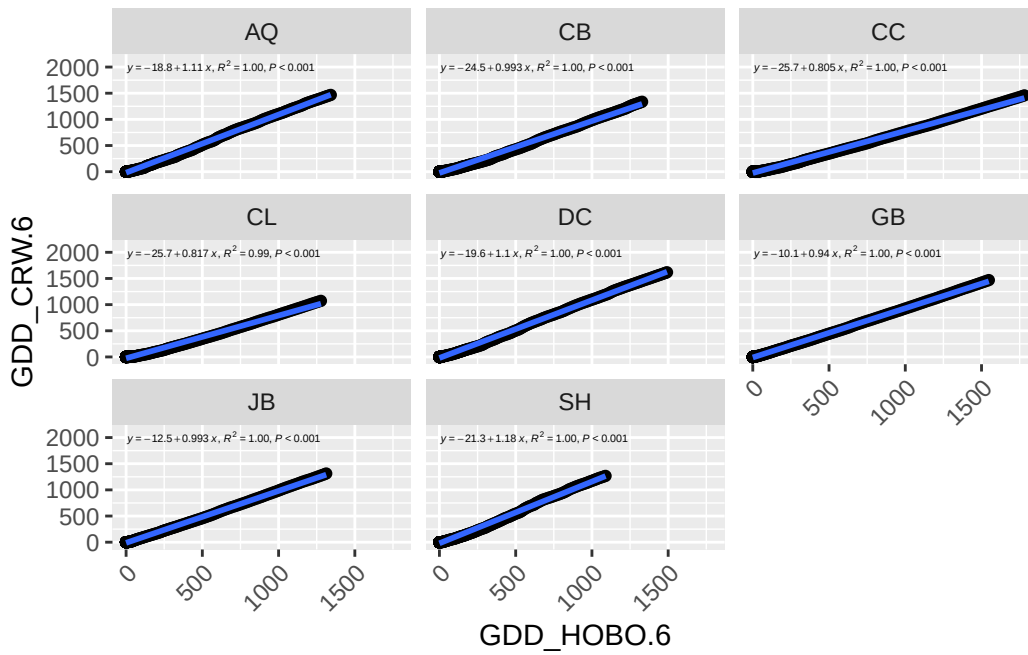
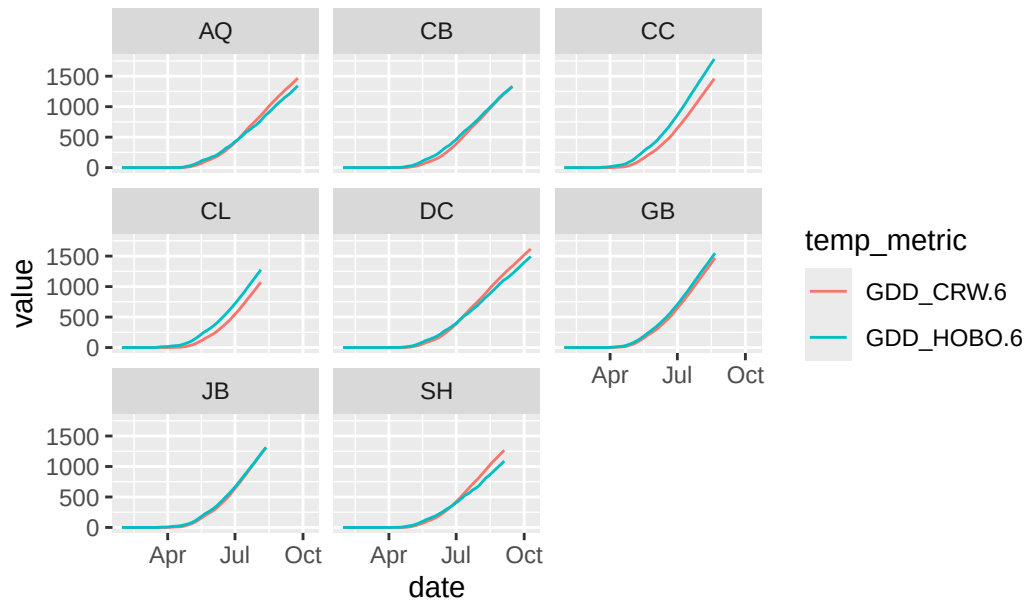


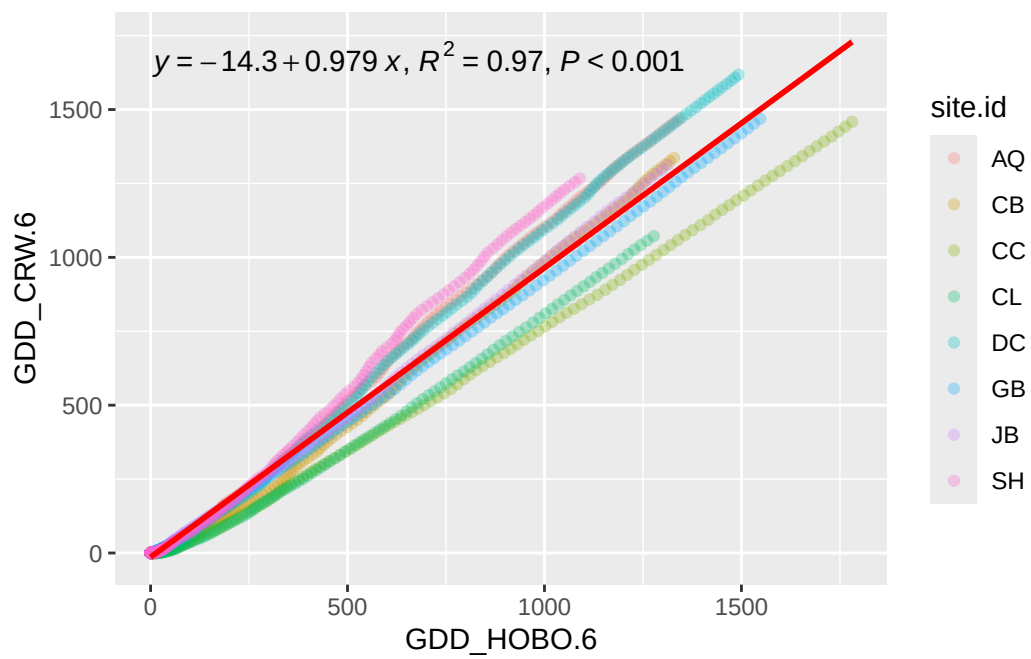
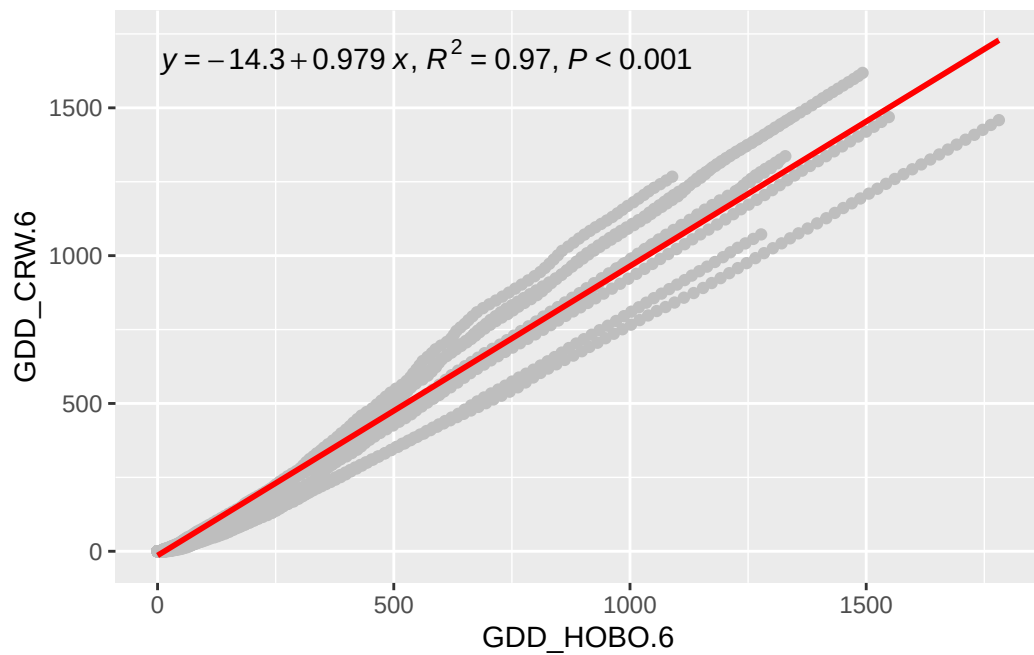
Tbase = 4 °C



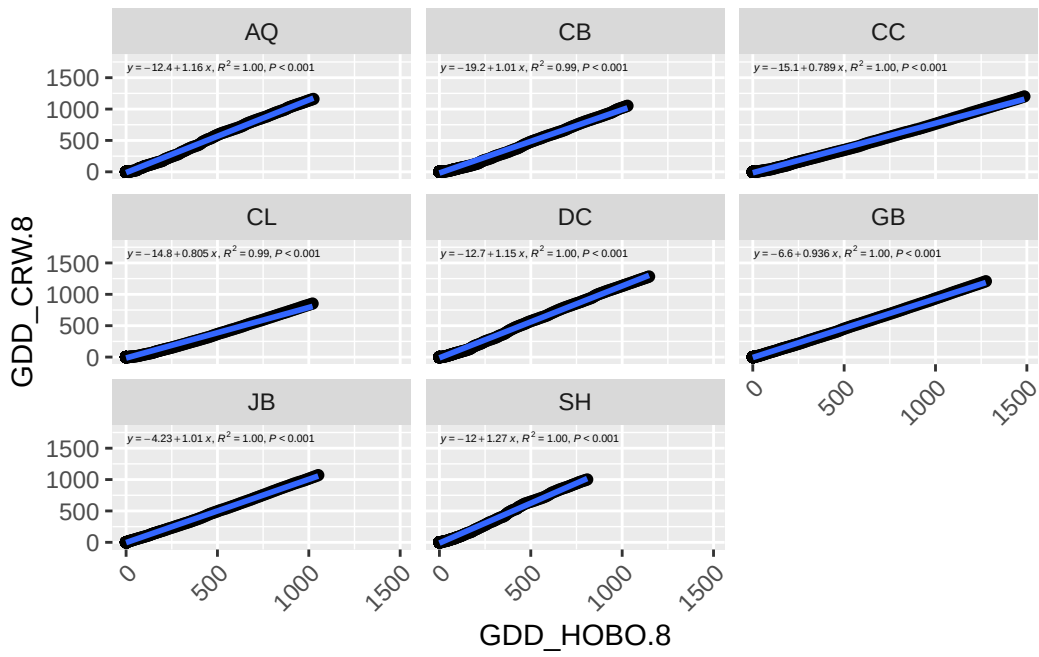
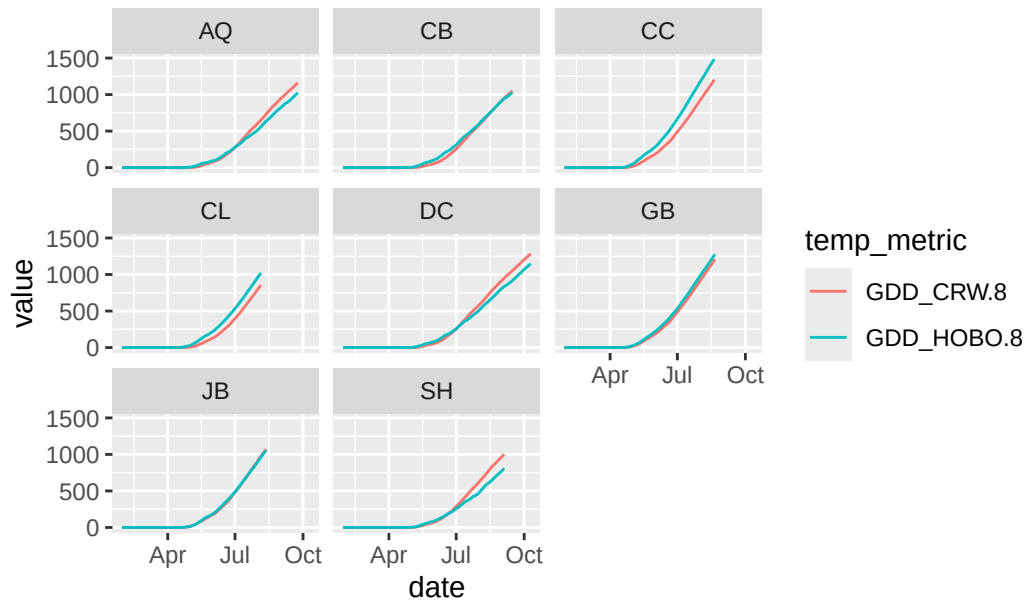


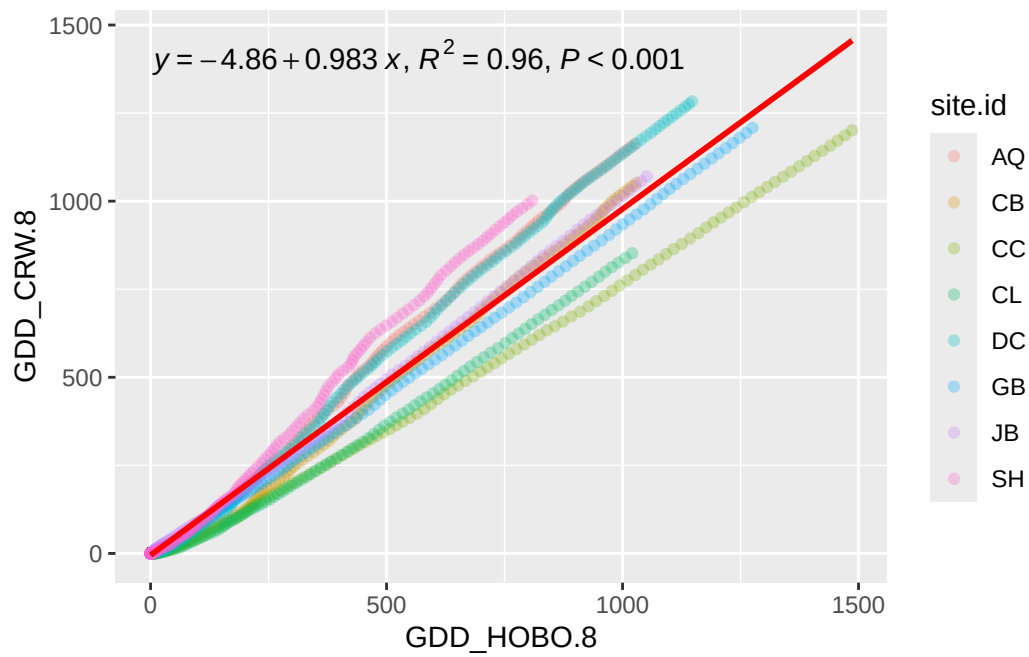
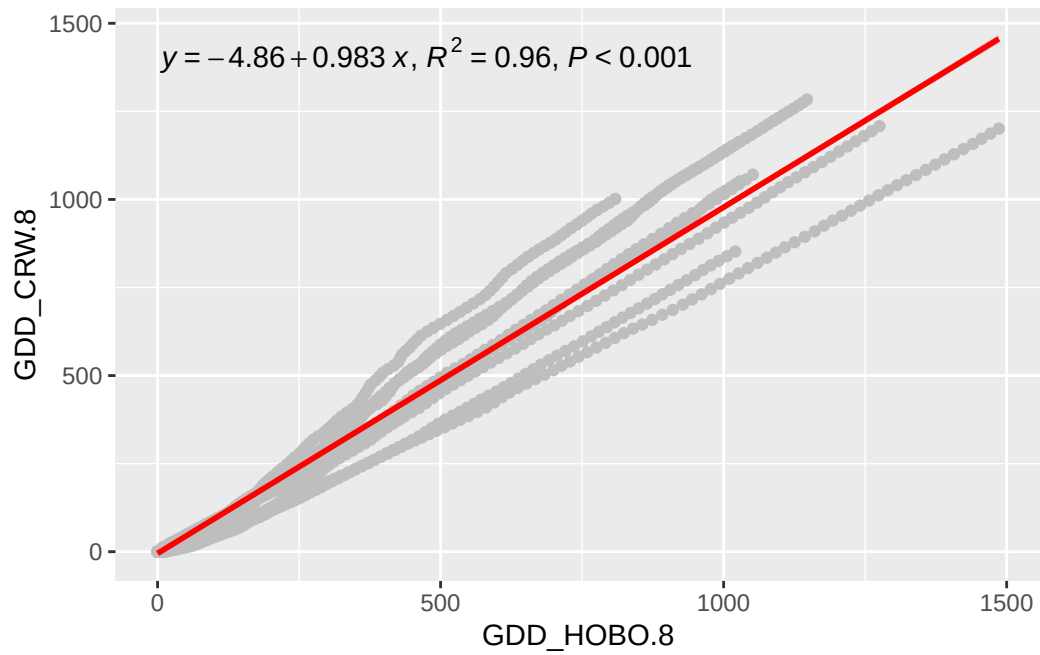
Tbase = 6 °C





Tbase = 8 °C





Compare HOBO and NASA GHR SST satellite data

Compare Growing Degree Days (GDD) - NASA GHR SST